Impact of Consensus Issuance Yield Curve Changes on Competitive Dynamics in the Ethereum Validator Ecosystem

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Abstract

This study analyzes competitive dynamics, elasticities, and event responses within the Ethereum validator ecosystem. Using historical blockchain data from the Beacon Chain genesis to May 2024, we examine differences in competitive advantages between validator subgroups under current and proposed issuance curves. We find that large validator pools (100+ validators) currently have 12-15% higher mean returns compared to solo validators, with proposed issuance reductions potentially exacerbating this advantage.

Elasticity analysis reveals solo stakers are highly sensitive to relative yield decreases compared to other staking categories, while being less responsive to absolute yield changes or DeFi yield fluctuations. Event studies show significant validator behavior changes in response to major ecosystem developments like withdrawal enabling and Bitcoin ETF launches, but limited impact from new staking protocols or macroeconomic events.

Our findings highlight the importance of addressing execution reward imbalances and maintaining relative yield competitiveness across validator segments to preserve a decentralized validator set. We conclude that a balanced approach combining MEV-Burn implementation with gradual issuance adjustments may best maintain a healthy, diverse validator ecosystem. Future research directions are proposed to further explore technological and alternative staking model impacts on validator behavior.

Contents

1. Introduction

1.1. Background and motivation

The current consensus issuance of Ethereum rewards is proportional to the square root of the deposit size. On top of these rewards, the block proposer is rewarded with execution rewards that depend on how much block builders are paying the proposer to include the block they propose as the next block. The Ethereum community has discussed changing the issuance curve in the Electra upgrade happening later this year, e.g. [1, 2, 3, 4]. Some of the most widely discussed curve candidates include a reward curve with gradual reward reduction proposed by Anders, a reward curve with substantial reward reduction proposed by Anders [4] and a reward curve with an economically capped reward issuance proposed by Vitalik [5].

We aim to contribute to the discussion by performing an empirical analysis on the current difference in competitive advantages between stakers to understand the root causes of the difference in competitive advantages and what these differences would look like if a different issuance curve was adopted today. In addition we perform an empirical analysis on the elasticity of different stakers to identify how likely their stake is to change if issuance and other variables change by analyzing the change in the number of active validators or validator exits in response to a change in influencing factor. In addition, we perform empirical event studies to analyze whether different stakers have changed their stake due to certain events by performing a two-sample t-test between an event window and the overall population.

1.2. Main Research Questions

While there has been discussion on various aspects of the issuance curve change there is limited understanding of how the competitive advantages among stakers might change if a different issuance curve was adopted. Additionally, the elasticity of stakers' behavior in response to changes in issuance and the impact of specific events on their staking patterns are not well-documented, highlighting a need for a comprehensive analysis of these dynamics.

We analyze Ethereum validator yields and behavior using historical blockchain data from the Beacon Chain genesis to May 2024 and pose the following main research questions:

- Q1. What are the quantitative differences in yield between validator pool sizes under the current and proposed issuance curves?
- Q2. How sensitive are validator deposits and exits to changes in staking APY?
- Q3. What factors influence the elasticity of stakers' stake adjustments?
- Q4. How do different validator categories respond to major ecosystem events?

Through these analyses, we aim to determine whether proposed issuance curve changes, discussed in question 1, would disproportionately impact smaller validators and potentially reduce decentralization of the network.

1.3. Findings

The study reveals notable competitive disparities among Ethereum validators, with large pools currently enjoying a 12% higher mean return compared to single-validator pools. This advantage could potentially increase to 13-15% under proposed issuance curve reductions without MEV-Burn implementation. Economically capped curves might create scenarios where large pools remain profitable while smaller ones incur losses.

Elasticity analysis shows solo stakers are particularly sensitive to relative yield decreases, emphasizing the importance of maintaining a balanced competitive landscape. This underscores the potential vulnerability of individual stakers to changes in the ecosystem's dynamics.

Event studies indicate varied validator responses to significant market occurrences. While some events had minimal impact, others elicited substantial reactions. The Ethereum Shanghai/Capella upgrade led to increased deposits for larger pools and higher exit levels for certain categories. The launch of the first Bitcoin ETF triggered significant validator exits across most pool sizes and categories, except for the largest pools and staking pools.

These findings highlight the complex interplay of factors influencing validator behavior, including yield competitiveness, liquidity considerations, and market events. The study emphasizes the need for a balanced approach to future protocol changes to maintain a healthy, decentralized validator ecosystem in Ethereum.

1.4. Limitations

Questions affected	**Limitation**	**Potential Impact**
Q2: Validator sensitivity to APY changes, Q3: Elasticity factors, Q4: Event study analysis	Time-lagged relationships not captured	May miss delayed effects between variables
Q1: Yield differences across validator pool sizes, Q2, Q3, Q4	Omission of potential third variables	Could miss confounding factors affecting both primary variables
Q1, Q2, Q3	Assumes constant validator count per entity	Under/overestimates relative yields for entities with changing validator numbers
Q1, Q2, Q3	Assumes no validator slashing	Minor yield inaccuracies across different operator types
Q2, Q3, Q4	Linear relationship assumed in elasticity model	May not capture non-linear behaviors or short-term fluctuations
Q4	Limited by fixed estimation windows	Could underestimate event impacts when information affects behavior before occurrence

Table 1: Limitations of the study

The study has a limitation in the comparison of data points that may inherently have a time-lagged relationship. When comparing two data points, the effect of a change in one variable might influence the other variable with a certain delay. However, for this study, data points are compared only at identical times. This approach does not fully account for the potential delayed impact that one data point might have on another.

The study does not consider the potential influence of unconsidered third variables on the relationship between the two primary variables being compared. In our analysis, we compare two variables directly, assuming a direct correlation or causation between them. However, it is possible that a third variable, not included in our study, may affect both primary variables, thereby confounding our results. This omission can lead to inaccurate conclusions regarding the nature and strength of the relationship between the two variables under investigation.
The yield analysis assumes the number of validators an entity has stays constant over the observation period. If the number of validators in an entity has significantly increased during the observation period the relative yield compared to other participants will be lower than the actual relationship. Similarly, the number of validators in an entity has significantly decreased during the observation period the relative yield compared to other participants will be higher than the actual relationship.

The yield analysis assumes that stakers are not slashed. This assumption does not hold in reality and thus there is minor inaccuracy in the results. In reality, validators get slashed for different reasons, for example, big institutions might get slashed when they are optimizing their setup for maximal execution rewards and small operators might get slashed for poor staking setups.

The elasticity model assumes a direct and linear relationship between influencing factors and validator behavior. Real-world behaviors might be influenced by additional, unmodeled factors leading to non-linear responses. Also, the normalization of validator exits and rewards to a daily level, while necessary for comparison, may obscure short-term fluctuations and finer-grained behaviors.

The event study utilizes an estimation window before and after an event to determine whether an event has changed the willingness to stake for certain groups. This methodology might not be accurate in estimating the effect of an event in certain cases. For example, an announcement of an event happening in the future might already affect willingness to stake before the actual event as stakers gain new information about the future that they can leverage when making their decision on staking. As a result, in such a scenario our event study might underestimate the effect an event has on stakers.

1.5. Structure of the study

The rest of the study is organized as follows. Section 2 outlines our data and methodology. Section 3 presents results and analysis of validator yields, issuance curves, elasticity, and event impacts. Section 4 discusses the implications of our findings. Section 5 concludes and suggests future research directions.

2. Data and methodology

This section describes the data we have used, how we have processed it, and the different methodologies we use. The first subsection describes the data collection and data processing. The second subsection describes the methodology used for analyzing the current competitive landscape and different curve candidates. The third subsection presents the methodology used for analyzing elasticity. The fourth subsection presents the methodology used in event studies.

2.1. Data and data processing

To systematically analyze the Ethereum validator ecosystem, we utilize multiple data sources as shown in Table 2. The table maps our data requirements, starting with validator labeling needs common to all analyses, followed by specific data requirements for each research question.

Research Question	**Primary Data Sources**	**Key Variables**
Validator labeling and categorization for all questions	Rated Network (Historical Beacon Chain data)	Validator activations and exits
	Rated Network (Validator metadata)	Labels, public keys, associated addresses
	Dune	Labels
	Etherscan	Labels
Q1: Yield differences across validator pool sizes	Rated Network (Historical Beacon Chain data)	Rewards, active validators
Q2: Validator sensitivity to APY changes	Rated Network (Historical Beacon Chain data)	Staking APY, validator activations/exits
Q3: Elasticity factors	Rated Network (Historical Beacon Chain data)	Staking APY, validator activations/exits
	Dune	Curve ETH-stETH liquidity APY
	TradingStrategy.ai	Aave V3 APY on WETH and wstETH
	CoinGecko	ETH price
Q4: Event study analysis	Rated Network (Historical Beacon Chain data)	Validator activations/exits

Table 2: Data sources

2.1.1. Data collection

Our dataset spans from the Beacon Chain genesis to slot 8,984,512 (May-02-2024 08:22:47 AM UTC) and comprises historical beacon chain data, validator metadata, DeFi data, and Ethereum price data.

Historical Beacon Chain Data

We acquire historical beacon chain data from Rated Network, encompassing deposits, withdrawals, validator activations, validator exits, and rewards. The rewards are further categorized into attestation, consensus, and execution rewards, providing a granular view of validator incentives. This dataset allows us to track validator performance metrics, staking ratios, and distribution patterns over time.

Validator Metadata

To accurately identify validators and label entities within the beacon chain, we collect data from Rated Network, Dune, and Etherscan. These sources provide the following:

- Rated Network: Metadata for validators, including validator public keys, validator indices, entity labels, activation and exit epochs, and deposit and withdrawal addresses. This enables precise identification of individual validators.
- Dune: Entity labels that enhance our understanding of validator affiliations and behaviors. Identification logic like deposit address factory contracts for a more complete labeling. These are used to supplement the Rated Network metadata.
- Etherscan: Supplementary entity labels to ensure comprehensive coverage and accuracy in our dataset.

By integrating these sources, we achieve a robust labeling system that facilitates the analysis of validator groupings and their respective characteristics.

DeFi Data

We obtain DeFi data to analyze validator behavior under various economic conditions from Dune and TradingStrategy.ai. This data includes the following:

- Dune: Data on Curve's ETH-stETH liquidity APY, offering a view of DeFi yield influence on staking decisions.
- TradingStrategy.ai: Historical liquidity returns for Aave v3 on WETH and wstETH, providing insights into lending market dynamics and their impact on validator behavior.

Ethereum Price

We obtain historical Ethereum market price data from CoinGecko as it offers an aggregate price from a variety of sources. This data enables the study of price movements and their effects on validator behavior.

2.1.2. Data processing

This subsection outlines our processing methodology for validator metadata, validator activations and exits, validator yield data, and DeFi yield data.

Validator metadata

Rated's validator metadata dataset serves as the foundation for our validator information. Entity categories are obtained from Dune and further supplemented through online searches. Validators are grouped into entities based on their shared withdrawal addresses; in cases where withdrawal addresses are not available, deposit addresses are utilized as an alternative. This methodology allows for the determination of the count of validators associated with each entity, allowing for an analysis of entity-specific validator distributions.

Validator activations and exits

By leveraging activation and exit epochs from Rated's validator metadata, we compute the historical counts of active validators and validator exits. These values are subsequently grouped according to entity validator count, category, and individual entity, based on the previously processed validator metadata.

For elasticity calculations, validator exits are normalized by dividing by the current active validators within their respective groups. This approach enables relative comparisons across groups with varying validator counts. Furthermore, both historical active validators and exits are normalized to a daily level to ensure alignment with the DeFi data.

Validator yield data

Rated's reward data is processed on a daily level. Slot and epoch data are integrated to ensure compatibility with other datasets. Validator rewards are then categorized by entity size, category, and individual entity.

To compute daily historical validator APYs for elasticity analysis, raw reward values are adjusted for decimal precision, divided by the corresponding validator stake, and subsequently annualized.

DeFi yield data

Aave liquidity APR values are averaged daily for WETH and wstETH pools, maintaining consistency with the daily frequency of most datasets. Additionally, daily percentage changes in the ETH market price are incorporated to enable elasticity analysis. Slots are added to all DeFi yield datasets, including the Curve ETH/stETH liquidity APR data, to ensure compatibility and integration with other datasets.

2.2. Current competitive landscape and comparing curve candidates

2.2.1. Current competitive landscape

The current consensus issuance of Ethereum rewards follows the equation $Y_i = cF\sqrt{D}$ with $c \approx 2.6$, F being 64, and D being the deposit size [1]. Under this curve, the consensus reward issuance is proportional to the square root of the deposit site and the issuance yield under the curve follows the equation $y_i = \frac{Y_i}{D} = \frac{cF}{\sqrt{D}}$. On top of these rewards block proposer is rewarded with execution rewards that, under the current MEV-Boost implementation, depend on how much block builders are paying the proposer to include the block they propose as the next block.

There are differences between the amount of consensus and execution rewards different staking entities achieve. Differences in consensus rewards are caused by how frequently entities are capable of restaking their rewards and how well they perform their validator duties. Differences in execution rewards are caused by differences in how much block builders are willing to pay for a block. Simplifying the block builders' behaviors, their bids depend on present CEX-DEX arbitrage opportunities and other available orderflow. Sophisticated validators can achieve higher expected returns from execution rewards than unsophisticated validators, for example by intentionally delaying the end of block auction to increase the expected value from CEX-DEX arbitrage opportunities resulting in higher expected return.

We analyze the current differences between stakers to determine how big competitive advantage different stakers have due to differences in consensus and execution rewards.

2.2.2. Curve candidates

Anders proposed multiple candidates for a new reward curve [4]. We focus on two curve candidates that are both determined as $Y_i = \frac{cF\sqrt{D}}{1+D/k}$ with $k = 2^{25}$ in the first candidate and $k = 2^{26}$ in the second.

In addition to curve candidates proposed by Anders, we analyze a curve candidate discussed by Vitalik that is economically capped [5]. The curve's issuance is determined as $Y_i = cFD(\frac{1}{\sqrt{D}} - \frac{0.5}{\sqrt{2^{25}-D}})$. In this curve, the consensus yield becomes negative around 26.8 million Ethereum limiting the feasibility of staking Ethereum.

We analyze what the impact of adopting different issuance curve candidates would look like for different validator groups. To do so we analyze the differences in consensus and execution rewards between different validator groups under the current consensus issuance curve over the last million slots of our dataset. Based on the differences we interpret why different validator groups have different historical returns. After determining the root causes of differences between validator groups we extrapolate our results and analyze how changing the issuance curve would reinforce or weaken the differences between different groups.

2.3. Elasticity

This subsection outlines the methodology used for analyzing the elasticity of validators in the Ethereum ecosystem. We analyze how strongly validators react, compared to other validators, to various factors that might influence their behavior including changes in:

- Validator Yield: This is a primary motivator for validator participation and will be significantly affected by the chosen reward structure.
- Yield Difference Between Entities: Stakers might strategically adjust their participation based on discrepancies in yield, seeking to maximize their returns. Analyzing these differences helps us understand competitive dynamics and validator mobility within the network.
- DeFi Yields: DeFi platforms offer alternative investment opportunities for validators. When DeFi yields are high, validators might be incentivized to move their funds to liquid staking solutions or other DeFi products, potentially leading to validator churn.

- Ethereum Dollar Price: The price of Ethereum directly impacts validator profitability as rewards are typically denominated in ETH. Validators also carry the risk of price fluctuations for their staked ETH, potentially influencing decisions to sell their stake and exit the network.

To analyze elasticity, we group validators, validator exits, and rewards based on three criteria:

- Entity Validator Count: This allows us to isolate the behavior of solo and small stakers and compare them against larger, institutional validators. Entities with a higher validator count might exhibit different elasticity patterns due to economies of scale or risk management strategies.
- Entity Category: Grouping validators by category such as exchange, liquid staking, and solo staker can reveal behavioral differences based on the entities' core business and risk tolerance.
- Individual Entity: Analyzing individual entities provides the most granular view and allows for the identification of specific actors who might be particularly sensitive to changes in the factors mentioned above.

By examining validator behavior across these groups, we can gain insights into how the chosen reward structure and overall economic conditions might influence validator participation in the Ethereum ecosystem.

Elasticity in this context is defined as the percentage change in the number of active validators or validator exits as a percentage of active validators in response to a percent change in the influencing factor. The formula used for calculating elasticity (E) is as follows:

$E = \frac{\%\Delta V}{\%\Delta X}$ where $\%\Delta V$ represents the percentage change in the number of validators and $\%\Delta X$ represents the percentage change in the influencing factor (Validator Yield, Yield Difference, DeFi Yields, or Ethereum Dollar Price). The test conducted is a two-sample t-test with a null hypothesis of no elasticity difference compared to the whole dataset.

2.4. Event studies

This subsection outlines the methodology used for event studies to analyze the impact of events on the staking behavior of validators in the Ethereum ecosystem. We examine how strongly different validator groups react to events, compared to other validators.

Event studies measure if an event has affected the behavior of a staker group more or less compared to all other stakers. We determine the timing slot when the event takes place and measure validator deposits and, with a Bitcoin ETF launch, the validator exits hourly for one week (50,400 slots) before and after the event slot. Validators are grouped based on two criteria: Entity Validator Count and Entity Category.

We calculate mean deposits/exits during the event window for each group and conduct a two-sample t-test against the overall population mean. Statistically significant differences indicate the event uniquely influenced that group's staking behavior compared to all validators.

3. Results and analysis

This section presents the results of different analyses. The first subsection discusses the results of analyzing the current competitive landscape and different curve candidates. The second subsection discusses the results of elasticity analysis. The third subsection discusses the results of the event studies.

3.1. Current competitive landscape and comparing curve candidates

3.1.1. Current competitive landscape

As shown in Table A1 there are significant differences in the mean consensus and the mean execution rewards as well as the combination of the two between different groups formed based on pool size, category, and entity. In Panel A the consensus rewards are significantly lower than the sample mean at 1% significance for all groups except pools with 100+ validators which are higher than the sample mean at 1% significance. The consensus rewards increase as the pool size increases. The mean consensus rewards for pools with size of 1 are 3.32%, 3.55% for pools with size of 100+, and 3.54% for the whole data.

For different pool sizes, the mean execution rewards are only significantly different than the mean for pools with 1 validator being lower than the mean of 0.63% at 5% significance. The mean execution rewards for pools with 1 validator are also significantly lower than the mean execution rewards of pools with 100+ validators of 0.87%. The median of execution rewards increases as the pool size increases.

Combining the consensus and execution rewards results in significantly lower than mean rewards for pools with 1 and 6-19 validators. Pools with size 1 have 0.48% percentage points lower total rewards compared to pools with size of 100+ meaning the mean rewards for pools with size of 100+ are 12% higher. Figure 1 illustrates these differences in rewards of differently sized pools at different points of the current issuance curve assuming MEV-burn is not implemented.

In Panel B the sample is split into groups by different categories of stakers. When grouped by category liquid restaking and solo stakers have significantly lower than mean consensus rewards of 3.49% and 3.50%, respectively, while CEX, liquid staking, and staking pools have significantly higher than mean consensus rewards of 3.55%, 3.56%, and 3.55%, respectively. Considering execution rewards no category's execution rewards statistically significantly differ from the sample mean. Combining both consensus and execution rewards, no category has statistically significantly different rewards from the mean.

Panel C observes the 10 biggest individual entities. Out of the entities Lido, Binance, Kraken, and OKX have significantly higher than mean consensus rewards of 3.56%, 3.60%, 3.56%, and 3.59%, respectively, while Coinbase, Rocketpool, Bitcoin Suisse, and Ether.Fi have significantly lower than mean consensus rewards of 3.53%, 3.47%, 3.53%, and 3.47%, respectively. No individual entity has execution rewards statistically significantly different from the mean. Combining consensus and execution rewards Lido has significantly higher than average rewards of 4.44% at 10% significance while Rockerpool has significantly lower than average rewards of 4.30% at 10% significance.

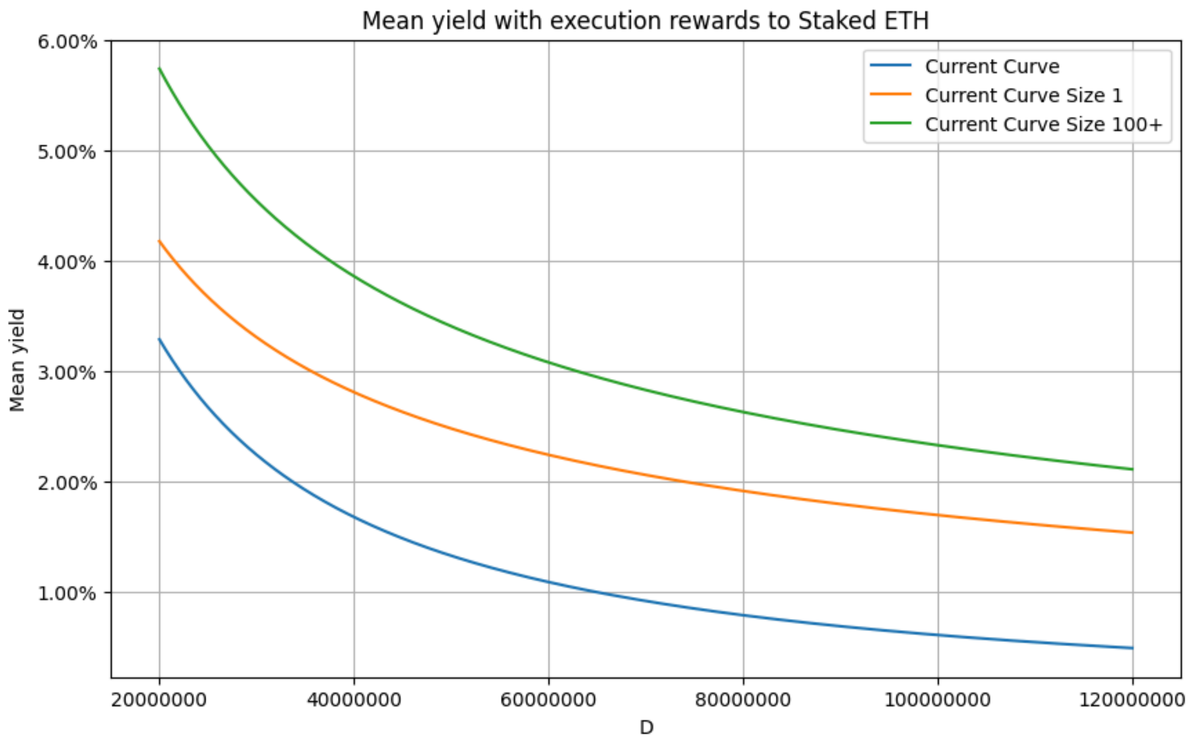

Figure 1: Mean yield of current issuance curve reduction by pool size with execution rewards

Under the current issuance curve, the consensus rewards increase as the pool size increases. We believe this is caused by an increase in the number of compounding periods as the pool size increases resulting in higher APY. Similarly, the median of consensus rewards increases as the pool size increases.

The median of execution rewards increases as the pool size increases. This indicates that the smallest pools rake smaller execution rewards than the largest pools. It is also important to note that when considering pools on the individual level the expected execution rewards are smaller for pools with few validators compared to bigger pools due to the variability of rewards as discussed by Anders [8].

A combination of both inferior consensus rewards and inferior execution rewards leads to a situation where the total rewards APY of our sample are 12% lower for pools with a size of 1 compared to pools with a size of 100+.

3.1.2. Gradual reward curve reduction

Anders proposed a reward curve with gradual reward reduction where the yield of consensus rewards follows a formula $y_i = \frac{cF}{\sqrt{D}(1+D/k)}$ and $k = 2^{26}$. The decrease in the mean rewards with and without execution rewards is visualized in Figures 2 and A1. The decrease in the median rewards with execution rewards is visualized in Figure A2.

Based on the analyzed sample if the issuance curve was to be changed and MEV-burn was not implemented this would result in a situation where staking pools with the size of 1 validator would have a mean APY of 3.13% while the pools with the size of 100+ validators would have a mean APY of 3.55% when considering the execution rewards. This means rewards for pools with a size of 100+

would be 13% higher than that of pools with a size of 1. Figure 3 illustrates these differences in rewards of differently sized pools at different points of the proposed issuance curve assuming MEV-burn is not implemented.

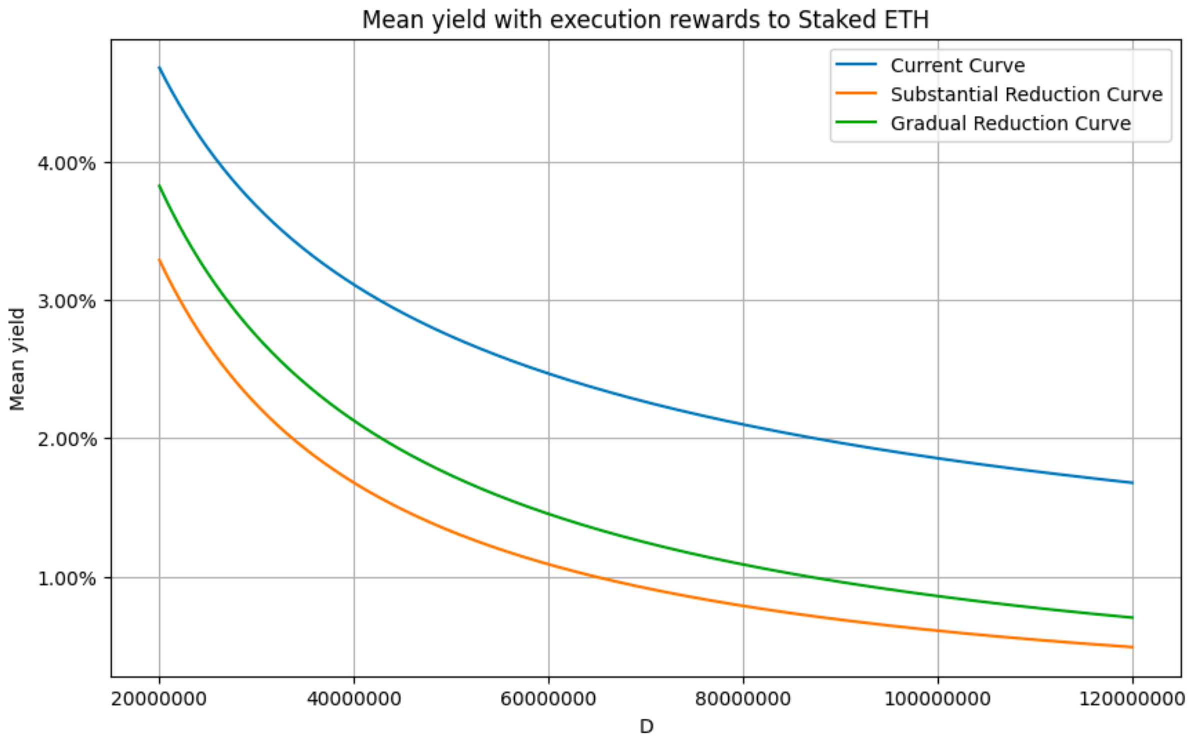

Figure 2: Mean yield of different issuance curves with execution rewards

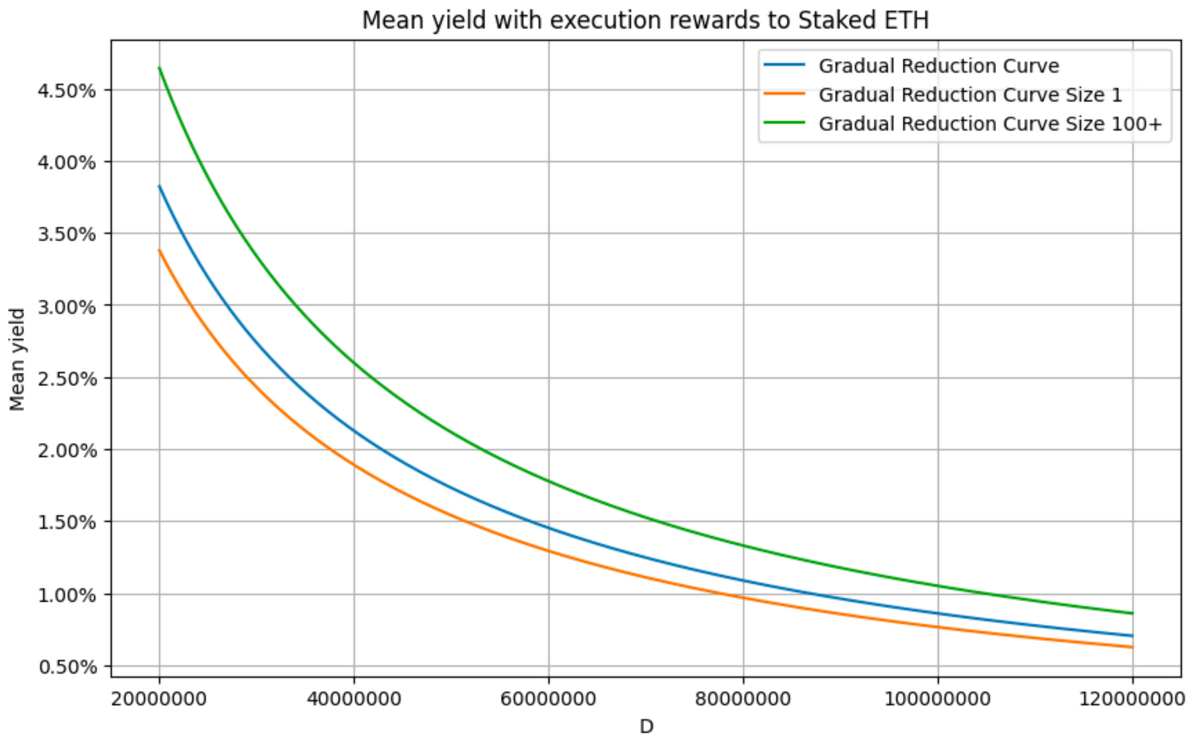

Figure 3: Mean yield of gradual issuance curve reduction by pool size with execution rewards

3.1.3. Substantial reward curve reduction

Anders also proposed an alternative reward curve with substantial reward reduction where the yield of consensus rewards follows a formula $y_i = \frac{cF}{\sqrt{D}(1+D/k)}$ and $k = 2^{25}$. The decrease in the mean rewards with and without execution rewards is visualized in Figures 2 and A1. The decrease in the median rewards with execution rewards is visualized in Figure A2.

Based on the analyzed sample if the issuance curve was to be changed and MEV-burn was not implemented this would result in a situation where staking pools with the size of 1 validator would have a mean APY of 2.63% while the pools with the size of 100+ validators would have a mean APY of 3.02% when considering the execution rewards. This means rewards for pools with a size of 100+ would be 15% higher than that of pools with a size of 1. Figure 4 illustrates these differences in rewards of differently sized pools at different points of the proposed issuance curve assuming MEV-burn is not implemented.

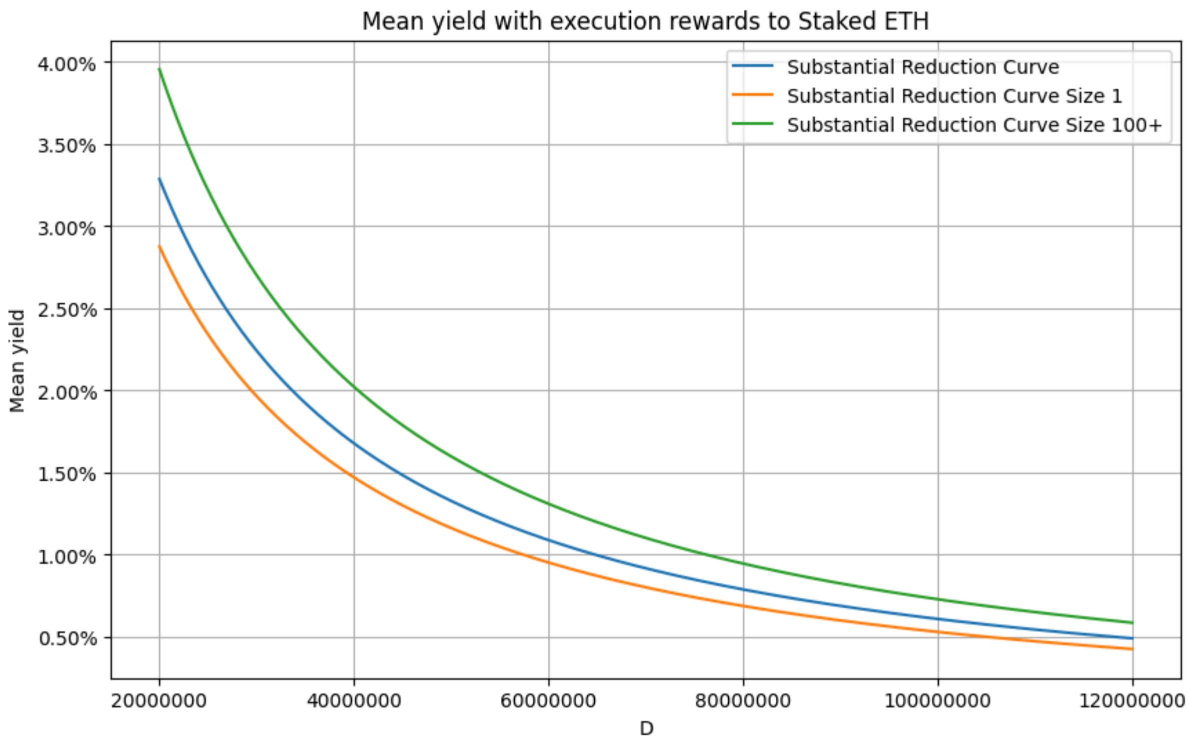

Figure 4: Mean yield of substantial issuance curve reduction by pool size with execution rewards

3.1.4. Economically capped curve

Vitalik proposed an economically capped reward issuance curve in his blog post. Under this curve, staking yield would follow $y_i = cF(\frac{1}{\sqrt{D}} - \frac{0.5}{\sqrt{2^{25} - D}})$ and become negative after a certain threshold staking amount is met. Figure 5 illustrates the economically capped curve compared to the current curve.

Based on the analyzed sample if the issuance curve were to be changed and MEV-burn was not implemented this would result in a situation where staking pools with the size of 1 validator would have a mean APY of 1.65% while the pools with the size of 100+ validators would have a mean APY of 1.97% when considering the execution rewards. Figure 6 illustrates these differences in rewards of differently sized pools at different points of the proposed issuance curve assuming MEV-burn is not implemented. In the figure, the smallest-sized pools have negative APY if the staked eth exceeds around 28,300,000 while the biggest-sized pools have negative APY if the staked eth exceeds around 28,700,000. This creates a scenario where the biggest pools could operate profitably while smaller pools would be losing money.

Figure 5: Mean yield of current and economically capped issuance curves with execution rewards

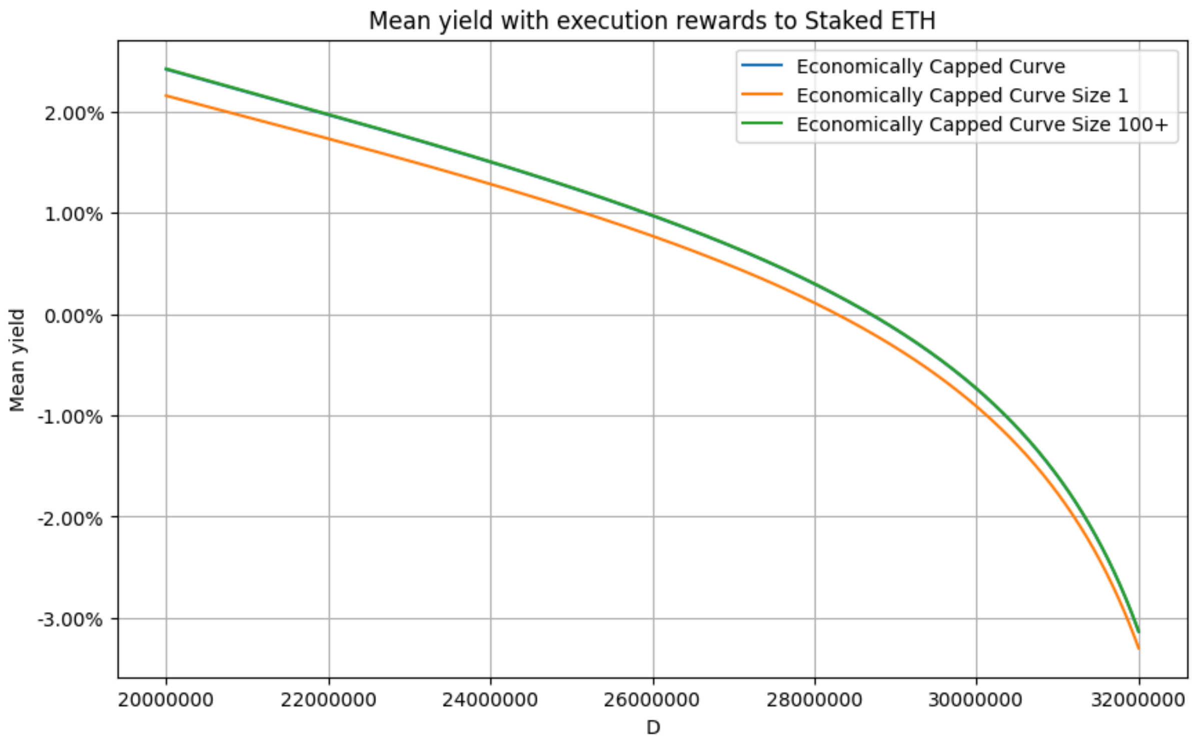

Figure 6: Mean yield of economically capped issuance curve reduction by pool size with execution rewards

3.2. Elasticity Analysis on Validator APY Changes

This section analyzes how sensitive different validator categories are to changes in staking APY, examining both deposits and exits across different staking categories. Table B1, visualized in Figure B1, presents elasticity estimates for validator behavior in response to APY changes between slots 6,840,000 and 8,985,900. Table B2, visualized in Figure B2, presents validator behavior in response to staking APY relative to other validators.

3.2.1. Pool Size Response to APY Changes

Among different pool sizes, we find distinct patterns of APY sensitivity. Pools with 2-5 validators and 6-19 validators demonstrate statistically significant negative elasticities (-0.4600 and -1.1733 respectively, both at 10% significance) for changes in total validator count. This suggests these smaller pools tend to reduce their validator count as staking APY rises.

Pools with 20-99 validators show different behavior when examining yield differences relative to larger pools. They exhibit a statistically significant positive elasticity (0.0616 at 10% significance) in total validator count when their APY differs from the largest pools (100+ validators). This suggests mid-sized pools tend to expand their validator set when they achieve higher relative yields compared to larger operations.

3.2.2. Category-Specific APY Sensitivity

Solo stakers demonstrate particularly notable sensitivity to relative yield differences. Their validator deposits show significant negative elasticity when comparing their yields to other staking categories:

- -0.9277 (5% significance) relative to liquid restaking
- -0.9153 (10% significance) relative to liquid staking
- -1.2740 (5% significance) relative to staking pools

Interestingly, solo stakers show no statistically significant elasticity when their yields differ from centralized exchanges (CEX), suggesting different competitive dynamics with this category.

3.2.3. Absolute vs. Relative Yield Sensitivity

When examining responses to absolute APY changes, we find limited significant effects across categories and individual entities. This contrasts with the stronger responses to relative yield differences, particularly among solo stakers and mid-sized pools. The lack of significant elasticity to absolute APY changes suggests stakers may be more sensitive to their competitive position relative to other staking options rather than to changes in the overall staking yield.

This analysis reveals that validator sensitivity to APY changes varies significantly by pool size and category, with particular importance placed on relative yield differences rather than absolute APY changes. These findings have important implications for understanding how changes in staking rewards might affect the distribution of validators across different categories.

3.3. Elasticity Analysis on External Market Factors

3.3.1. Validator ETH Price Elasticity

We investigate the sensitivity of validator behavior to changes in the Ethereum price by calculating elasticities for validator exits, deposits, and total validator count changes relative to the overall dataset. Table C1, visualized in Figure C1, presents the results for subgroups based on pool size, category, and individual entity using a two-sample t-test against a null hypothesis of no elasticity difference compared to the complete dataset.

When grouping by category, solo stakers exhibit a statistically significant negative elasticity of -0.4031 at 1% significance, indicating they are less elastic and less likely to exit their validators when Ethereum prices rise. Staking pools display a positive elasticity of 1.1772 at 10% significance, suggesting they are more elastic, with a higher propensity to exit validators as prices increase. The liquid restaking category has a negative elasticity of -6.2982 at 10% significance for total validator count changes, implying it is less elastic than the full dataset and tends to decrease its overall validator set during Ethereum price increases.

Among the largest individual entities, Lido and Binance demonstrate negative elasticities of -0.0585 at 5% significance and -0.7476 at 10% significance, respectively, suggesting they are less elastic and less likely to exit validators when Ethereum prices rise. OKX exhibits a statistically significant negative elasticity of -1.0009 at 1% significance for changes in total validator count, indicating it decreases its validator set when Ethereum prices increase. Conversely, Ledger Live and Mantle exhibit positive elasticities of 4.1524 at 10% significance and 6.8474 at 10% significance, respectively, indicating higher elasticity and an increased likelihood of exiting validators during Ethereum price increases relative to other stakers. Bitcoin Suisse shows a statistically significant positive elasticity of 4.0668 at 1% significance for changes in total validator count, suggesting an increase in its validator set as Ethereum prices rise.

3.3.2. Validator DeFi Yield Elasticity

Tables C2 and C3, visualized in Figures C2 and C3, present the elasticity estimates of validator exits, deposits, and total validator count changes in response to Aave v3 WETH supplier yield and Curve stETH-WETH liquidity yield between slots 6,206,400 and 8,985,900.

The analysis reveals no statistically significant effect of DeFi yield changes on staker behavior across deposits, exits, or total validator counts.

3.4. Event studies

3.4.1. Rocket Pool mainnet launch - 9th November 2021

The Rocket Pool mainnet launch on November 9th, 2021 introduced a new decentralized liquid staking protocol to the Ethereum ecosystem. [6]

Analyzing the impact of the Rocket Pool's mainnet launch, Table D1, visualized in Figure D1 reveals no statistically significant differences in validator deposits before and after the event. All of the pool sizes in Panel A, as well as the pool categories in Panel B, do exhibit a lack of robust statistical significance. These results suggest that the Rocket Pool's mainnet launch did not significantly influence the willingness to stake across the various pool classifications.

3.4.2. First 0.75% FED interest hike - 16th June 2022

The first major interest rate hike by the Federal Reserve on June 16th, 2022 was one of the largest single increases in decades, signaling a shift towards an aggressive tightening monetary policy cycle by the Fed to combat high inflation. This increased the opportunity cost of capital tied up in illiquid staking positions compared to other yield opportunities. It also applied downward pressure on risk assets like cryptocurrencies, affecting the risk/reward dynamics of staking returns.

Table D2, visualized in Figure D2, presents the results of the event study analyzing the impact of the first 0.75% FED interest rate hike on validator deposits. The results suggest that the first 0.75% FED interest rate hike did not significantly impact the willingness to stake, as evidenced by the lack of statistically significant differences in validator deposits before and after the event across the dataset and various pool categories.

3.4.3. Ethereum Staking Withdrawals Open - 12th April 2023

The enabling of Ethereum staking withdrawals through the Shanghai/Capella upgrade on April 16th, 2023 was significant for providing liquidity to previously inaccessible and illiquid staked ETH after over two years of lockup. It removed a major opportunity cost associated with staking by allowing validators to access and manage their staked capital and rewards more flexibly. The upgrade also marked meaningful progress in Ethereum's transition to proof-of-stake consensus, demonstrating advancement toward the network's final architecture. [7]

Table D3, visualized in Figure 7, shows statistically significant increases in mean validator deposits for the 20-99 validator pool size at 5% significance, the Solo Stakers category at 5% significance, and the CEX category at 1% significance after the upgrade. However, no significant changes were observed for other groups.

Complementing these findings, Figures 8 and 9 illustrate the cumulative validator exits across pool sizes and categories after the upgrade, respectively. These figures reveal higher exit levels for larger pool sizes of 20-99 and 100+ and certain categories like CEX and Liquid Staking.

The statistically significant increase in validator deposits for larger pool sizes, coupled with the observed higher exit levels post-upgrade, suggests the newfound ability to withdraw staked ETH

appears to have influenced the staking behavior of these pools, potentially due to the increased liquidity and flexibility in managing their staked capital and rewards.
However, the lack of significant differences for smaller pool sizes indicates that the impact of the upgrade on the willingness to stake was more pronounced for larger pools, while smaller pools and the overall staking ecosystem remained relatively unaffected.

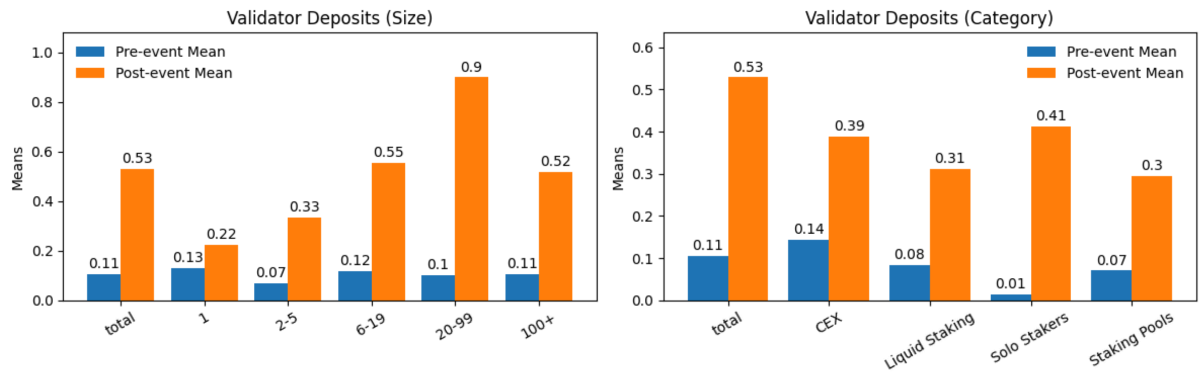

Figure 7: Validator deposits one week before and after the Ethereum Shanghai/Capella upgrade that allowed staker withdrawals. The deposits have been normalized by the net deposits of the subgroup.

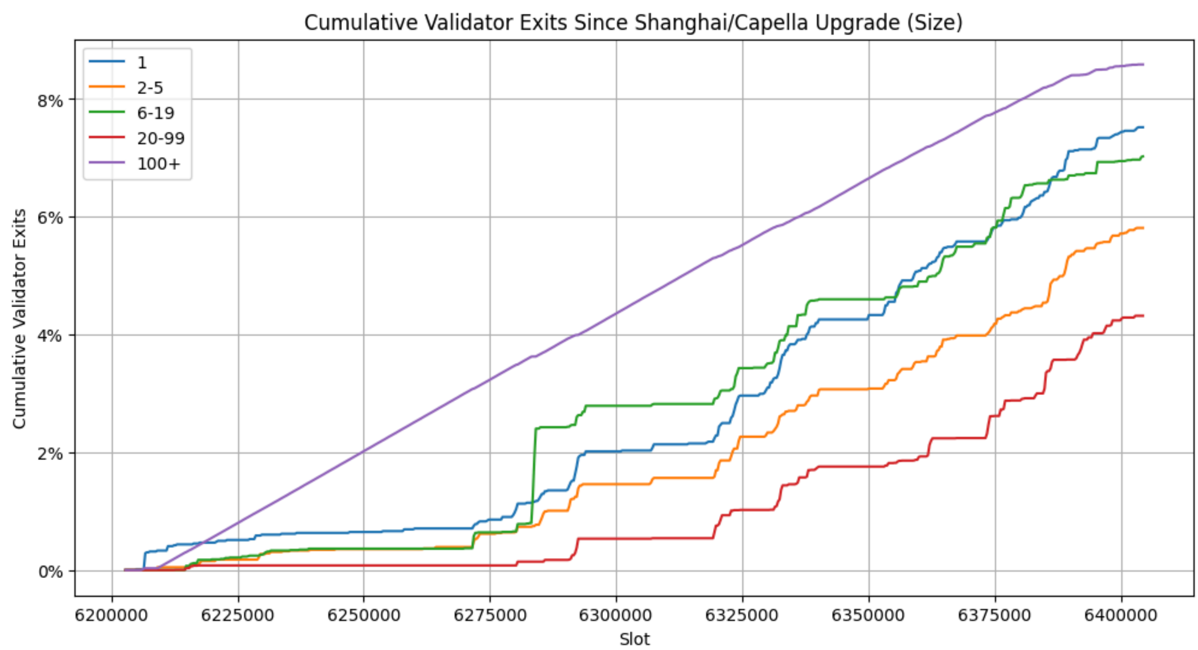

Figure 8: Cumulative validator exits by pool size for one month after the Shanghai/Capella upgrade. The exits have been normalized by the subgroup validator count at slot 6202800.

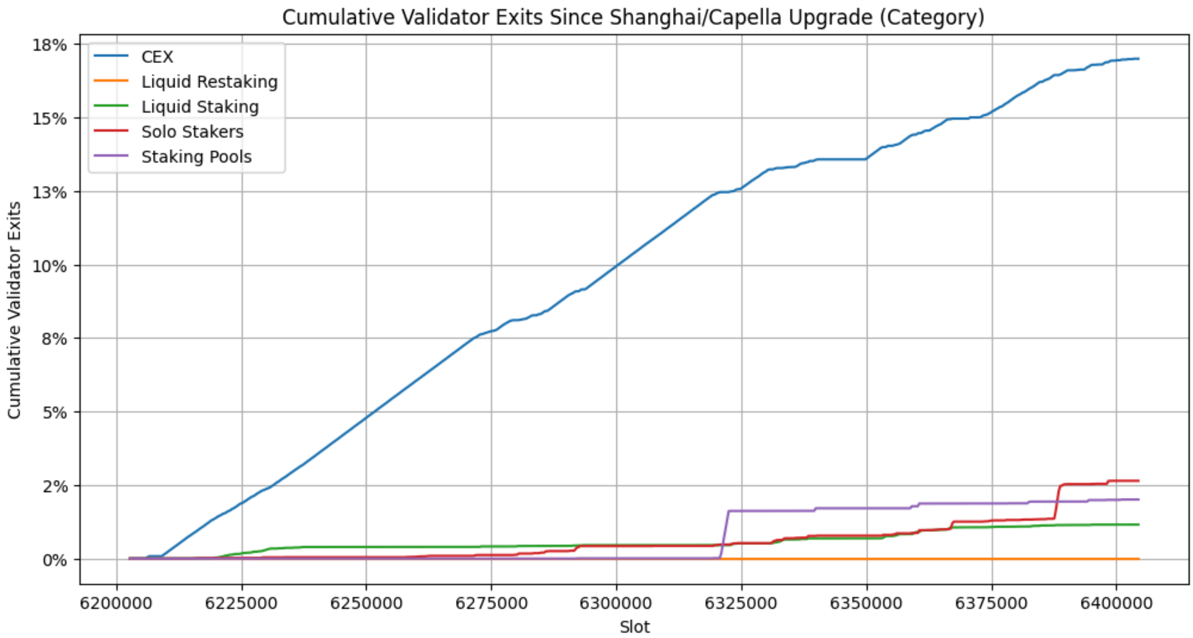

Figure 9: Cumulative validator exits by category for one month after the Shanghai/Capella upgrade. The exits have been normalized by the subgroup validator count at slot 6202800.

3.4.4. Bitcoin ETF Launch - 11th January 2024

As seen in Table D4, and visualized in Figures 10 and 11, the launch of the first Bitcoin ETF in the US on January 11th, 2024 marked a significant shift in mainstream cryptocurrency adoption and legitimacy. These regulated investment vehicles allowed investors to gain exposure to cryptocurrencies through conventional financial products, signifying their integration into traditional finance.

The Bitcoin ETF launch had a significant impact on validator exits across most pool sizes and categories. The 1, 2-5, and 6-19 validator pool sizes exhibited statistically significant increases in mean exits post-event at the 1% significance level. The 20-99 pool size showed a significant increase at the 5% level, while the 100+ pool size experienced a significant decrease in exits at the 1% level. Among categories, the CEX group had a significant increase in exits at the 1% level. The Liquid Staking and Solo Stakers categories also saw significant increases at the 5% level. Conversely, the Staking Pools category experienced a significant decrease in exits at the 1% level.

The event did not significantly affect validator deposits for most groups at the 10% significance level. The exceptions were the 100+ pool size with a significant increase at the 5% level, the Liquid Staking category with a significant decrease at the 10% level, and the Liquid Restaking category with a significant increase at the 5% level.

These results suggest the Bitcoin ETF launch primarily influenced validator exits, with most groups increasing exits, except larger pools and staking pools. Validator deposits remained largely unchanged, with only marginal effects observed for the 100+ pool size and Liquid Staking category.

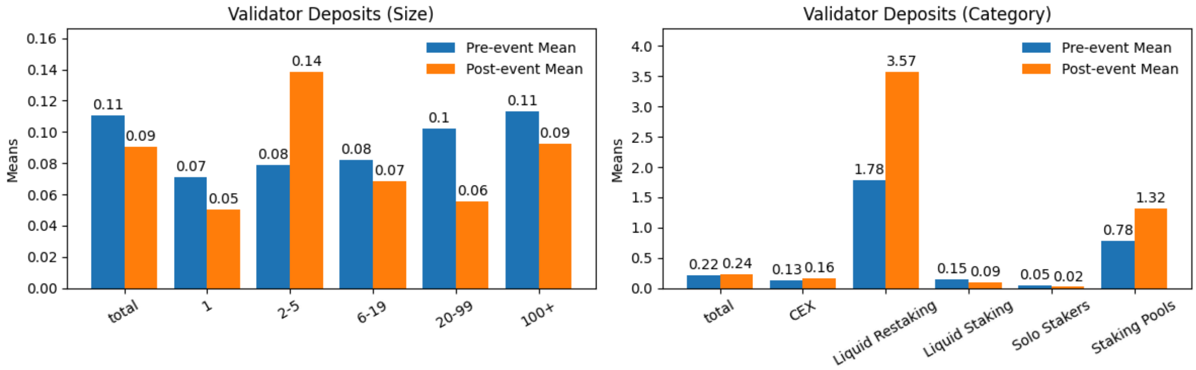

Figure 10: Validator deposits one week before and after the launch of the first Bitcoin ETF. The deposits have been normalized by the net deposits of the subgroup.

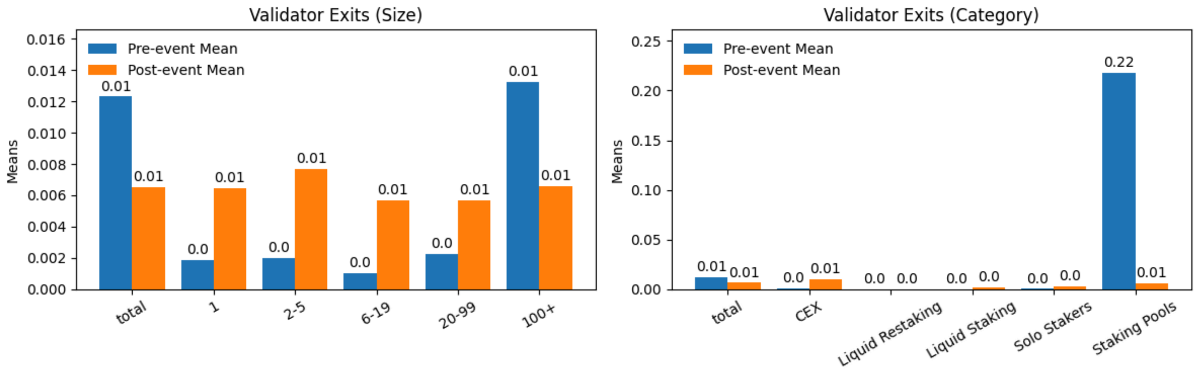

Figure 11: Validator exits one week before and after the launch of the first Bitcoin ETF. The exits have been normalized by the active validator count of the subgroup.

4. Discussion

Under the current issuance curve the larger validators pools with 100+ validators consistently outperform smaller pools and solo validators in terms of mean returns. This advantage can be attributed to better execution reward strategies and increased compounding efficiency. The new issuance curves proposed by Anders [8] would reduce the consensus rewards resulting in amplification of these advantages. Issuance curve with gradual reduction would result in an advantage of 13% and curve with substantial reduction would result in an advantage of 15%.

MEV-Burn has been widely discussed as an improvement that would decrease the importance of execution rewards for validators and the variance of execution rewards. We believe that in an ideal situation, MEV-Burn should be implemented in parallel or before a consensus reward reduction to minimize the competitive advantage the current execution rewards give to big pools. However, our analysis concludes that if the consensus rewards were decreased according to gradual or substantial consensus reward reduction curves before the implementation of MEV-Burn the competitive advantage big pools would get would be a few percentage points. On the other hand, this means that combining consensus rewards reduction even with an inefficient implementation of MEV-Burn that would, for example, burn half of the execution rewards would result in a smaller difference in percentwise total rewards between the big and small pools.

Adopting an economically capped curve proposed by Vitalik or some other economically capped issuance curve could lead to a situation where big pools operate at a profit while small pools are losing money. This could lead to a situation where big pools capture the majority of the staking market share pushing small pools out of the market leading to an oligopoly. This could happen even if MEV-Burn was highly effective as big pools could be willing to operate at a loss to gain market share. Additionally, even if the notional yield of big pools was negative some of them could still be willing to operate at a loss if they are charging a higher management fee for staking the assets.

The elasticity analysis revealed solo stakers exhibited highly statistically significant negative elasticity for validator deposits in response to decreases in their staking yield relative to other categories like liquid restaking, liquid staking, and staking pools. This suggests a correlation where solo stakers seem particularly sensitive to negative changes in their yield competitiveness compared to other staking vehicles. In contrast, changes in total staking yield and fluctuations in DeFi yields do not appear to have elicited such pronounced effects. This observation highlights the potential importance of balancing the relative staking yield differences among validator segments for solo stakers and smaller entities to maintain their portion of the validator set.

The findings indicate addressing execution reward imbalances is crucial before implementing major issuance changes. A balanced approach involving MEV-Burn and gradual issuance adjustments may better maintain a healthy, decentralized validator ecosystem, preserving incentives for diverse participation.

5. Conclusion

The purpose of this study is to analyze the current differences in competitive advantages between stakers and what these differences would look like if a different issuance curve was adopted today, the elasticity of different stakers to identify how likely their stake is to change if issuance or other variables change, and event studies to analyze whether different stakers have changed their stake due to certain events.

This study utilizes historical beacon chain, validator metadata, DeFi data, and Ethereum price data. The dataset spans from late 2020 to early 2024. We analyze differences in current competitive advantages between subgroups and find that pools with a size of 100+ have 12% higher mean returns than pools with a size of 1. Assuming MEV-Burn is not implemented, gradual and substantial issuance curve reductions proposed by Anders [4] would result in a competitive advantage of 13% and 15%, respectively. Due to these differences, if MEV-Burn was to be implemented and it was inefficient, the combination of the new consensus issuance curve and MEV-Burn would lead to a better situation where the competitive advantage of big pool operators compared to small pools would diminish. In addition, we want to point out that adopting an economically capped curve such as the one proposed by Vitalik [5] could lead to a situation where big pool operators are operating at a profit while small pools are operating at a loss due to differences in the ability to capture execution rewards.

We analyze the elasticity of stakers relative to issuance rates and different variables. We find that solo stakers are highly sensitive to decreases in their staking yield relative to other categories. This suggests a correlation where solo stakers appear to be more vulnerable to negative changes in their yield competitiveness compared to other staking vehicles. In contrast, changes in total staking yield and fluctuations in DeFi yields did not seem to elicit such pronounced effects.

Our study also considers different staker reactions to major events. The Rocket Pool mainnet launch and the FED's first major interest rate hike did not significantly impact validator deposits, suggesting new staking protocols and macroeconomic fluctuations have limited immediate influence on staking behavior. However, the Ethereum Shanghai/Capella upgrade enabling staking withdrawals resulted in increased deposits for larger pools and higher exit levels for certain categories like CEXs and Liquid Staking providers. This highlights the importance of liquidity and capital management flexibility in staking decisions for larger entities.

The launch of the first Bitcoin ETF significantly increased validator exits across most pool sizes and categories, except for the largest pools and staking pools. This behavior, prevalent in the retail-heavy CEX category, may reflect a "sell the news" phenomenon where stakers capitalize on positive news by exiting their positions.

In conclusion, this study provides insights into the competitive dynamics, elasticities, and event responses within the Ethereum validator ecosystem. The findings underscore the importance of addressing execution reward imbalances and balancing relative staking yield differences among validator segments. A balanced approach involving MEV-Burn and gradual issuance adjustments may better maintain a healthy, decentralized validator set, preserving incentives for diverse participation. Future research could explore the impact of regulatory changes, technological advancements, and alternative staking models on validator behavior, accounting for potential time lags and non-linear effects.

5.1. Future research directions

Our study considers notional yields. Future research could extend the analysis to consider real yields in addition to notional yields.

Our study analyzes direct and linear relationships between data points. Future research could also consider non-linear relationships and time lags in effects.

The analyzed events in the event studies are limited. Future research could consider additional events. These could include launches of staking service providers incentivizing stakers to stake with them by offering airdrops. In the future, the historical dataset for validator exits will grow and offer a new dimension to event studies in addition to deposits used in most of our events. In addition, future research could also analyze the impact of events based on announcements of events happening in the future instead of the time the event actually takes place.

For geographic decentralization of the validator set, investigating regional regulatory and taxation impacts could prove insightful. Evolving regulatory frameworks and taxation policies across jurisdictions may influence stakers' operational viability, profitability, and locational preferences.

Future research could also investigate the effects of upgrades or technological improvements to staking infrastructure, such as advancements in hardware or software that could impact operational efficiencies and costs for validators.

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[3] https://notes.ethereum.org/@mikeneuder/iiii
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[7] https://ethereum.org/en/staking/withdrawals/
[8] https://ethresear.ch/t/properties-of-issuance-level-consensus-incentives-and-variability-across-potential-reward-curves/18448

Appendices

A. Issuance Curves

Table A1

Current Issuance Curve

This table reports results on analysis of consensus and execution rewards and their differences between different entities between slots 7 984 513 and 8 984 512. Panel A reports mean, t-statistic of whether the subgroup's mean is different from dataset's mean, and median for whole dataset as well as by different pool sizes. Panel B reports the same values grouped by different categories. Panel C reports the same values for the 10 biggest pools. ***, **, and * denote statistical significance of two-sample t-test at the 1%, 5%, and 10% levels.

Panel A. Whole dataset and pools by size

	Consensus + Execution Rewards			Consensus Rewards			Execution Rewards			
Subgroup	Mean	t(Mean)	Median	Mean	t(Mean)	Median	Mean	t(Mean)	Median	N
Whole dataset	4.41%	-	4.07%	3.54%	-	3.67%	0.87%	-	0.37%	1000000
1	3.947%***	-4.59	3.95%	3.3156%***	-28.47	3.64%	0.6314%**	-2.34	0.27%	9699
2-5	4.39%	-0.2	3.97%	3.3996%***	-18.97	3.66%	0.99%	1.24	0.27%	10735
6-19	4.238%**	-2.39	4.00%	3.4673%***	-13.46	3.66%	0.77%	1.34	0.31%	19307
20-99	4.34%	-1.31	4.00%	3.5239%***	-5.43	3.66%	0.82%	-0.93	0.30%	46178
100+	4.42%	0.93	4.08%	3.5509%***	6.04	3.67%	0.87%	0.42	0.38%	914081

Panel B. Pools by category

	Consensus + Execution Rewards			Consensus Rewards			Execution Rewards			
Category	Mean	t(Mean)	Median	Mean	t(Mean)	Median	Mean	t(Mean)	Median	N
CEX	4.42%	0.58	4.09%	3.5505%***	3.75	3.67%	0.87%	0.27	0.40%	273776
Liquid Restaking	4.40%	-0.25	3.98%	3.4932%***	-10.98	3.59%	0.90%	0.52	0.36%	29205
Liquid Staking	4.43%	1.12	4.07%	3.5564%***	8.09	3.67%	0.88%	0.46	0.38%	357121
Solo Stakers	4.31%	-1.33	4.06%	3.5049%***	-6.28	3.69%	0.80%	-0.84	0.35%	16007
Staking Pools	4.46%	1.09	4.10%	3.5544%***	2.77	3.66%	0.91%	0.87	0.41%	46703

Panel C. Largest pools

	Consensus + Execution Rewards			Consensus Rewards			Execution Rewards			
Pool	Mean	t(Mean)	Median	Mean	t(Mean)	Median	Mean	t(Mean)	Median	N
Lido	4.444%*	1.65	4.07%	3.5642%***	12.65	3.67%	0.88%	0.64	0.38%	312012
Coinbase	4.42%	0.17	4.10%	3.5327%***	-5.25	3.67%	0.88%	0.6	0.40%	147345
Binance	4.47%	1.19	4.11%	3.5973%***	13.23	3.68%	0.87%	0.14	0.40%	39461
Rocketpool	4.2986%*	-1.86	4.05%	3.4655%***	-16.3	3.66%	0.83%	-0.57	0.36%	27308
Kraken	4.39%	-0.4	4.07%	3.5613%***	3.48	3.68%	0.83%	-0.67	0.37%	25731
OKX	4.53%	1.43	4.12%	3.5937%***	7.94	3.70%	0.93%	0.81	0.40%	15914
Bitcoin Suisse	4.40%	-0.11	4.10%	3.5265%***	-2.78	3.68%	0.88%	0.11	0.41%	15462
Ledger Live	4.52%	1.29	4.14%	3.55%	0.38	3.67%	0.97%	1.26	0.45%	14977
Ether.Fi	4.29%	-1.45	3.99%	3.4688%***	-11.53	3.59%	0.82%	-0.55	0.38%	14258
Mantle	4.40%	-0.08	4.05%	3.54%	-0.13	3.64%	0.86%	-0.07	0.39%	12962

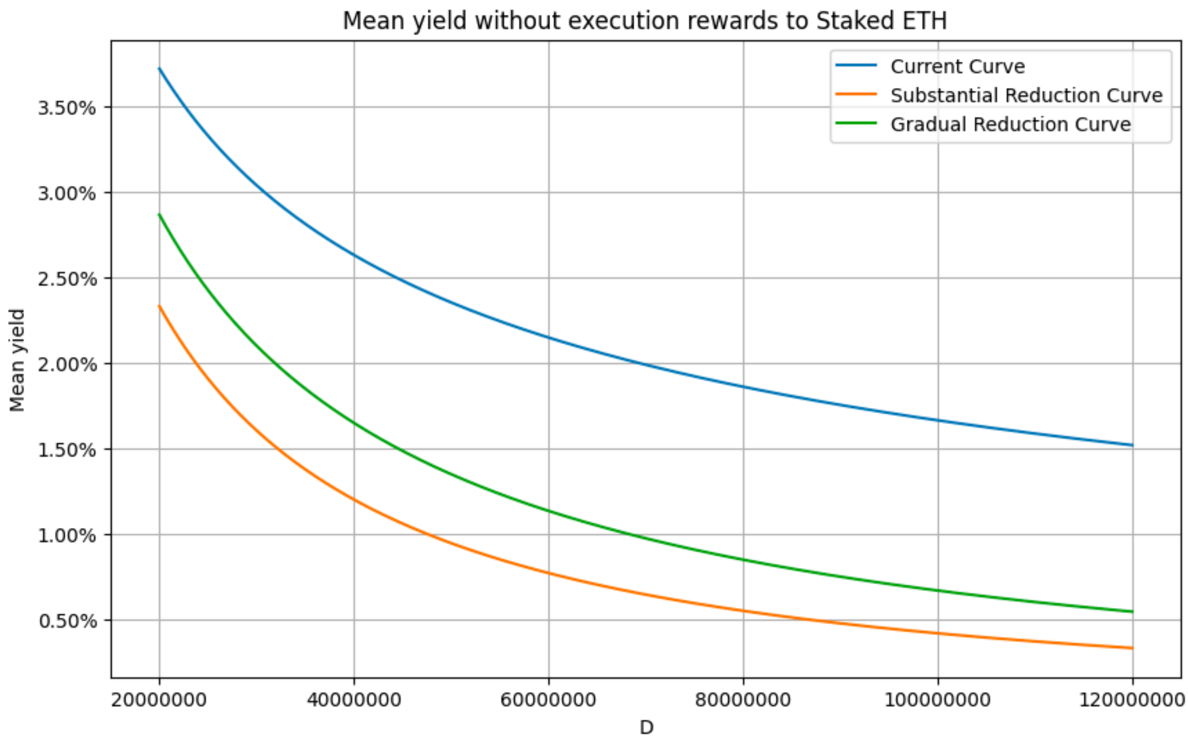

Figure A1: Mean yield of different issuance curves without execution rewards

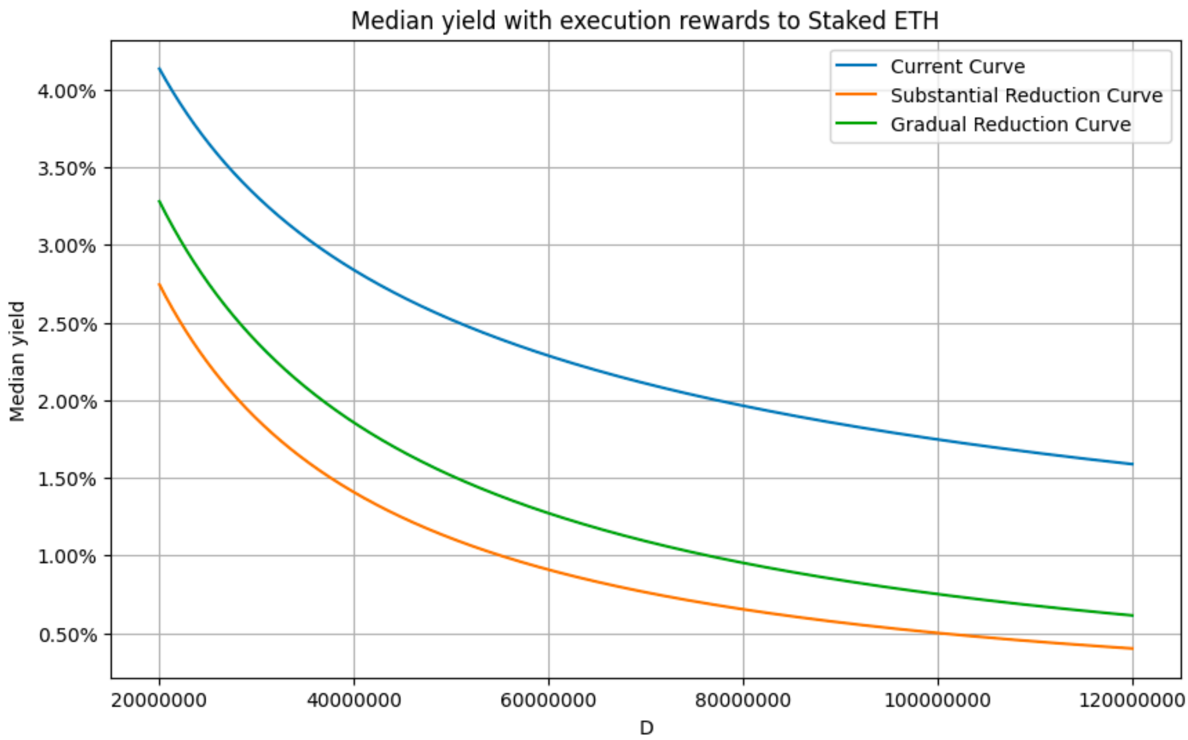

Figure A2: Median yield of different issuance curves with execution rewards

B. Staking APY Elasticity

Table B1

Staking APY Elasticity

This table reports elasticity estimates of validator exits, validator deposits, and the change in total validator count with respect to the staking APY daily between slots 6,840,000 and 8,985,600. Panel A reports the mean elasticity, t-static, and standard deviation for the whole dataset as well as by different pool sizes. Panel B reports the same values grouped by different categories of staking entities. Panel C reports the same values for the largest individual staking pools. ***, **, and * denote statistical significance of two-sample t-test at the 1%, 5%, and 10% levels against a null hypothesis of no elasticity difference compared to the whole dataset.

Panel A. Whole dataset and pools by size										
	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
Whole dataset	14.4308	-	218.4485	56.9506	-	685.5406	3.8742	-	42.8035	298
1	5.0082	-0.71	66.3333	-3.3040	-1.50	111.7696	-0.1657	-1.60	7.5863	298
2-5	12.4533	-0.10	253.6697	-1.1851	-1.44	129.013	-0.4600*	-1.68	12.089	298
6-19	-8.3294	-1.51	141.909	-4.8866	-1.43	295.6051	-1.1733*	-1.67	29.9475	298
20-99	106.5238	0.87	1810.773	74.5275	0.28	850.2656	-21.0260	-1.08	397.4911	298
100+	123.2935	0.69	2723.667	852.7969	1.00	13757.4837	-27.3605	-1.09	491.6845	298
Panel B. Pools by category										
	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
CEX	-12.9753	-0.45	560.8725	41.7387	1.52	370.1452	2.5641	1.08	20.8675	298
Liquid Restaking	0.4965	-0.16	25.6381	-189.6798	-1.31	2395.1657	-27.9477	-1.02	478.8957	298
Liquid Staking	-5.1982	-0.47	201.5205	-4.0196	0.04	287.5772	-0.8883	-0.67	25.3425	298
Solo Stakers	-8.3848	-0.81	127.9088	-1.6572	0.15	30.3316	1.2195	0.37	21.5389	298
Staking Pools	0.5033	-0.11	195.7062	-22.1302	-0.62	281.7663	0.1264	-0.20	19.8394	298
Panel C. Largest pools										
	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
Lido	-6.9421	0.57	178.3498	0.5916	0.40	262.9217	0.1009	0.32	12.1665	298
Coinbase	12.6800	0.83	500.8729	26.1315	0.60	992.6826	2.3591	1.06	33.5537	298
Binance	-15.6369	0.42	183.3799	4.8730	0.48	123.9306	1.6150	0.96	19.5617	298
Rocketpool	-6.1248	0.59	116.7748	0.3977	0.41	61.1904	1.3612	0.95	8.8483	298
Kraken	21.0061	0.99	437.8077	48.7953	1.00	633.8088	0.9299	0.63	23.3169	298
OKX	-9.5339	0.53	124.2035	-21.2453	0.08	272.3651	0.1488	0.36	5.9919	298
Bitcoin Suisse	9.6249	0.88	128.8742	1.9622	0.43	51.9266	-0.0048	0.27	11.6407	298
Ledger Live	36.5668	1.22	493.7663	39.6782	0.90	559.0886	0.0505	0.30	9.3323	298
Ether.Fi	-2.2131	0.67	23.7612	-506.3010	-0.87	8034.9361	0.6125	0.53	15.2424	217
Mantle	2.9227	0.76	54.2478	-29.1855	0.00	7194.892	-0.2379	0.15	12.0873	203
Other Stakers	21.3271	1.06	231.8739	138.3665	1.41	1676.2228	0.0246	0.20	37.1115	298

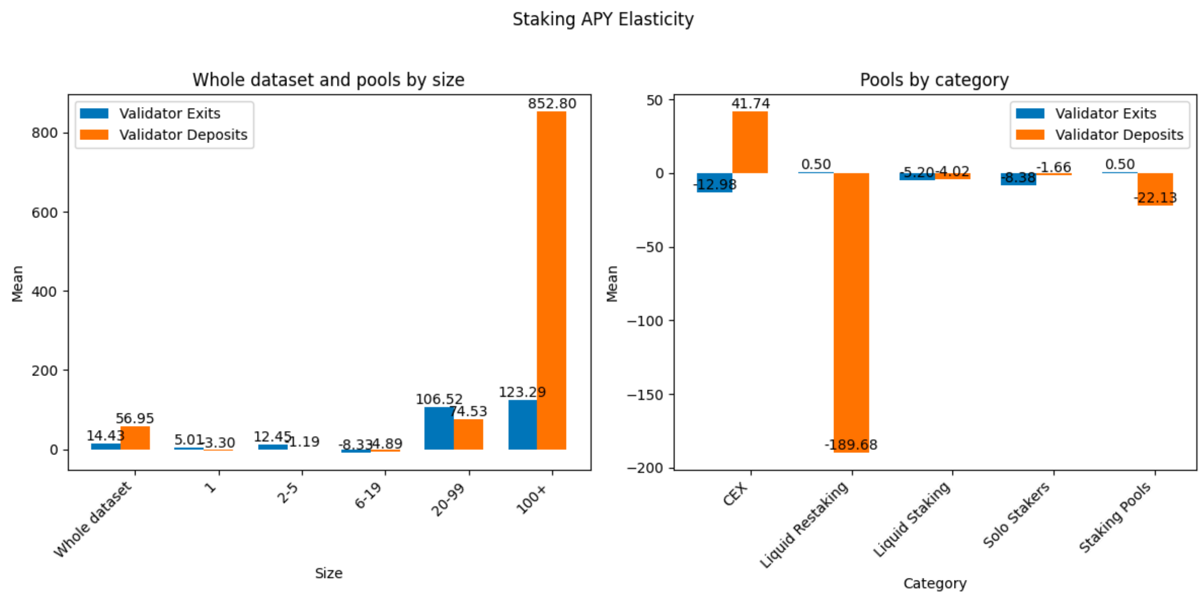

Figure B1: Validator exits and total count change relative to staking APY. The amounts are normalized relative to subgroups.

Table B2

Staking APY Difference Elasticity of Small and Solo Stakers

This table reports elasticity estimates of validator exits, validator deposits, and the change in total validator count with respect to the staking APY difference between stakers daily between slots 6,840,000 and 8,985,600. Panel A reports the mean elasticity, t-static, and standard deviation for the whole dataset as well as by different pool sizes. Panel B reports the same values grouped by different categories of staking entities. Panel C reports the same values for the largest individual staking pools. ***, **, and * denote statistical significance of one-sample t-test at the 1%, 5%, and 10% levels against a null hypothesis of no elasticity difference compared to the whole dataset.

Panel A. Pools by size compared to 100+										
	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
1	0.8184	1.20	11.7467	69.2844	1.29	928.9801	0.004	0.16	0.4305	298
2-5	0.0629	0.77	1.4042	2.9616	0.79	64.7791	0.0014	0.15	0.156	298
6-19	0.1686	0.49	5.9964	-14.039	-1.08	223.8343	-0.0068	-0.58	0.202	298
20-99	-0.195	-0.82	4.1246	-5.9817	-0.80	129.7086	0.0616*	1.89	0.561	298
Panel B. Solo Stakers compared to other categories										
	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
CEX	0.1204	1.24	1.676	-0.9661	-1.16	14.3917	-0.0006	-0.11	0.0922	298
Liquid Restaking	-0.0774	-1.22	1.0938	-0.9277**	-2.27	7.0447	-0.0032	-0.86	0.064	298
Liquid Staking	0.2681	1.39	3.3198	-0.9153*	-1.87	8.4538	-0.0057	-0.80	0.1228	298
Staking Pools	0.3922	0.95	7.1359	-1.274**	-2.35	9.3765	-0.0136	-1.08	0.2186	298

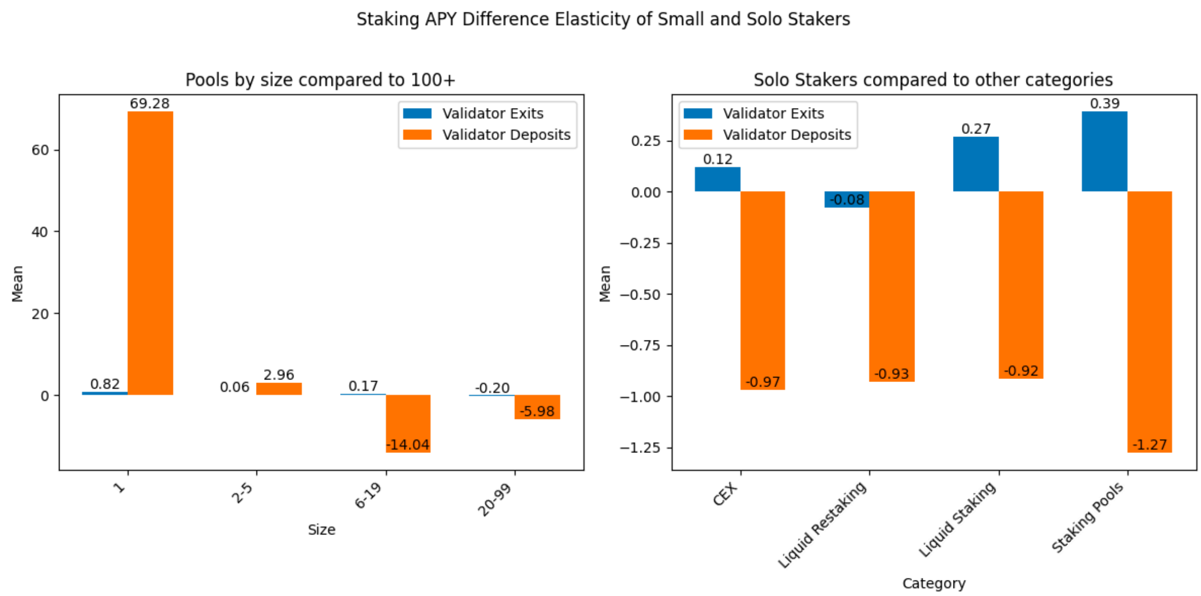

Figure B2: Validator exits and total count change relative to staking APY difference of small and solo stakers. The amounts are normalized relative to subgroups.

C. Elasticity for External Market Factors

Table C1

Ethereum Price Elasticity

This table reports elasticity estimates of validator exits, validator deposits, and the change in total validator count with respect to the price of Ethereum hourly between slots 6,206,400 and 8,985,900. Panel A reports the mean elasticity, t-static, and standard deviation for the whole dataset as well as by different pool sizes. Panel B reports the same values grouped by different categories of staking entities. Panel C reports the same values for the largest individual staking pools. ***, **, and * denote statistical significance of two-sample t-test at the 1%, 5%, and 10% levels against a null hypothesis of no elasticity difference compared to the whole dataset.

Panel A. Whole dataset and pools by size										
	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
Whole dataset	0.096	-	6.2555	0.0001	-	0.1859	-0.0275	-	10.6454	9266
1	-0.0166	-0.95	9.589	0.0005	0.18	0.1892	-0.0949	-0.22	27.4506	9266
2-5	-0.0541	-1.18	10.5836	0.0022	0.71	0.2129	-0.1582	-0.56	19.653	9266
6-19	0.0704	-0.16	14.5326	0.0012	0.32	0.2821	0.1464	0.56	27.824	9266
20-99	0.158	0.40	13.5522	-0.0015	-0.51	0.2217	0.2849	0.85	33.5914	9266
100+	0.0959	0.00	6.6641	0.0001	0.00	0.1932	-0.0427	-0.09	11.289	9266
Panel B. Pools by category										
	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
CEX	0.1351	0.26	13.0568	0.0023	0.69	0.2536	0.1432	0.70	20.9308	9266
Liquid Restaking	0.5032	1.41	27.0534	-0.0802	-0.79	9.8377	-6.2982*	-1.83	329.957	9266
Liquid Staking	0.0036	-1.12	4.939	-0.0021	-0.76	0.2054	-0.2414	-1.16	14.282	9266
Solo Stakers	-0.4031***	-2.78	16.1427	-0.0003	-0.11	0.2366	0.3272	1.55	19.2289	9266
Staking Pools	1.1772*	1.81	57.1452	-0.0006	-0.14	0.4081	0.0617	0.12	71.6029	9266
Panel C. Largest pools										
	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
Lido	-0.0585**	-1.99	4.1272	-0.0039	-1.40	0.2002	-0.1621	-0.70	15.0589	9266
Coinbase	0.1259	0.29	7.5226	0.0045	1.02	0.3757	0.0886	0.40	25.5882	9267
Binance	-0.7476*	-1.92	41.7772	-0.0011	-0.18	0.6107	0.6435	1.45	43.26	9268
Rocketpool	0.1379	0.22	17.62	-0.0001	-0.04	0.2341	-0.3758	-0.72	45.0988	9269
Kraken	0.2409	0.71	18.4889	0.0047	0.93	0.4438	-0.0744	-0.15	27.7324	9270
OKX	-0.0626	-0.72	20.19	0.0062	0.85	0.6754	-1.0009***	-2.57	34.9165	9271
Bitcoin Suisse	-0.0177*	-1.70	1.4897	-0.0014	-0.62	0.1191	4.0668***	3.00	131.111	9272
Ledger Live	4.1524*	1.96	199.5568	0.0094	1.06	0.8268	-3.3666	-1.46	219.6608	9273
Ether.Fi	7.0115	1.61	310.6214	0.3372	1.30	18.804	-25.0225	-0.18	9893.341	5247
Mantle	6.8474*	1.68	281.1932	-0.0154	-0.06	17.2844	-7.2857	-0.33	1553.384	4887
Other Stakers	0.0594	-0.33	8.8241	0.0013	0.36	0.2848	0.0412	0.29	20.3249	9276

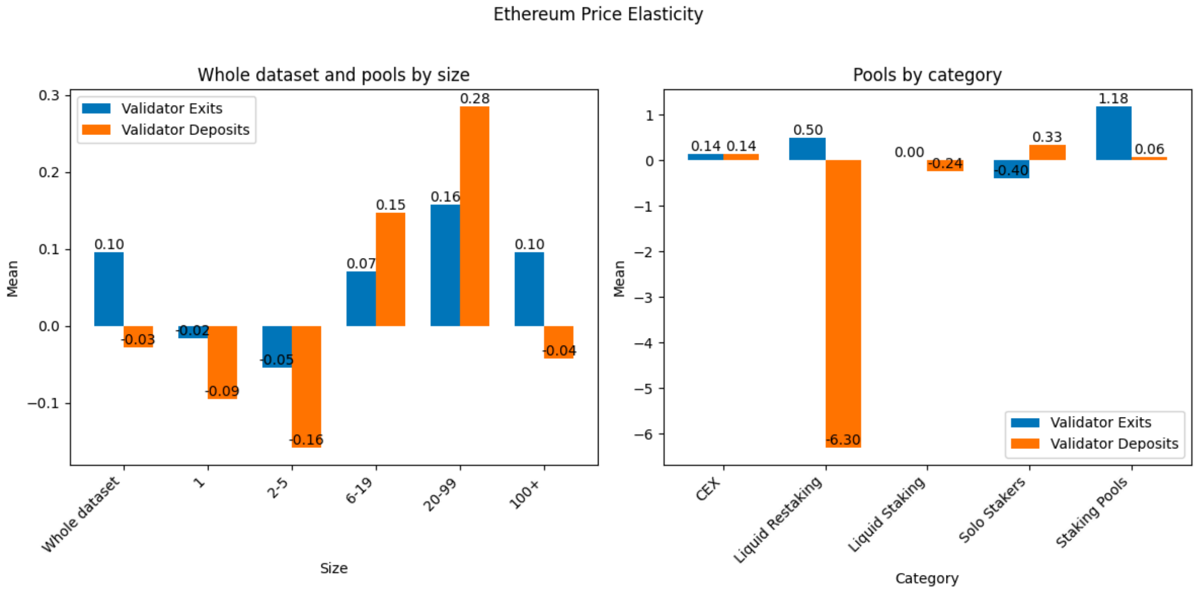

Figure C1: Validator exits and total count change relative to Ethereum price. The amounts are normalized relative to subgroups.

Table C2

Aave v3 WETH Supply APR Elasticity

This table reports elasticity estimates of validator exits, validator deposits, and the change in total validator count with respect to the Aave v3 WETH Supply APR daily between slots 6,206,400 and 8,985,600. Panel A reports the mean elasticity, t-static, and standard deviation for the whole dataset as well as by different pool sizes. Panel B reports the same values grouped by different categories of staking entities. Panel C reports the same values for the largest individual staking pools. ***, **, and * denote statistical significance of two-sample t-test at the 1%, 5%, and 10% levels against a null hypothesis of no elasticity difference compared to the whole dataset.

Panel A. Whole dataset and pools by size

	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
Whole dataset	-0.2077	-	5.7005	0.0051	-	0.1428	-0.0473	-	4.0027	387
1	-0.1536	0.17	2.9008	0.0014	-0.50	0.0306	0.2537	1.09	3.6612	387
2-5	0.1275	0.81	5.8543	-0.0019	-0.92	0.0394	-0.4217	-0.79	8.4595	387
6-19	0.379	1.34	6.4486	0.0021	-0.37	0.0691	-0.5132	-0.98	8.4703	387
20-99	-0.2984	-0.27	3.2713	0.0016	-0.41	0.0844	0.1252	0.52	5.1453	387
100+	-0.2214	-0.03	6.1878	0.0055	0.04	0.1523	-0.0427	0.02	4.4229	387

Panel B. Pools by category

	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
CEX	-1.0142	-0.77	19.8695	0.005	0.00	0.2562	0.736	0.73	20.7499	387
Liquid Restaking	-0.0201	0.65	0.3956	0.0042	-0.02	1.1935	0.2612	0.22	27.5972	387
Liquid Staking	0.029	0.69	3.5453	0.0037	-0.18	0.0551	0.1547	0.45	7.9588	387
Solo Stakers	-0.3923	-0.40	7.196	0.0029	-0.29	0.0439	0.2121	0.60	7.4822	387
Staking Pools	0.652	0.68	24.3983	0.016	0.83	0.2142	-1.0075	-0.76	24.6376	387

Panel C. Largest pools

	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
Lido	0.1727	1.14	3.2087	0.0037	-0.18	0.0576	0.1771	0.47	8.3925	387
Coinbase	-0.0481	0.54	1.2338	0.0054	0.01	0.3752	-0.3808	-0.58	10.4889	387
Binance	-6.7885	-0.99	130.431	0.0028	-0.29	0.0556	6.7009	1.02	130.4586	387
Rocketpool	-0.0959	0.33	3.5718	0.0103	0.36	0.2444	-0.1068	-0.17	5.6388	387
Kraken	0.2371	1.13	5.2252	-0.0117	-1.21	0.2321	0.413	0.42	21.2743	387
OKX	-2.999	-0.99	55.2635	0.0021	-0.40	0.0341	3.4799	1.25	55.5155	387
Bitcoin Suisse	0.0183	0.77	0.8265	-0.0002	-0.72	0.004	-0.2636	-0.59	6.0359	387
Ledger Live	-0.1581	0.17	1.3094	0.0241	0.75	0.48	-0.3288	-0.45	11.5039	387
Ether.Fi	0	0.72	0	0.1674	0.98	2.447	-1	-0.57	24.5197	219
Mantle	7.9208	1.00	115.633	-0.0003	-0.74	0.0126	-8.0955	-0.99	116.2026	204
Other Stakers	0.1449	0.90	5.1361	0.0066	0.12	0.2174	-0.882	-1.57	9.6589	387

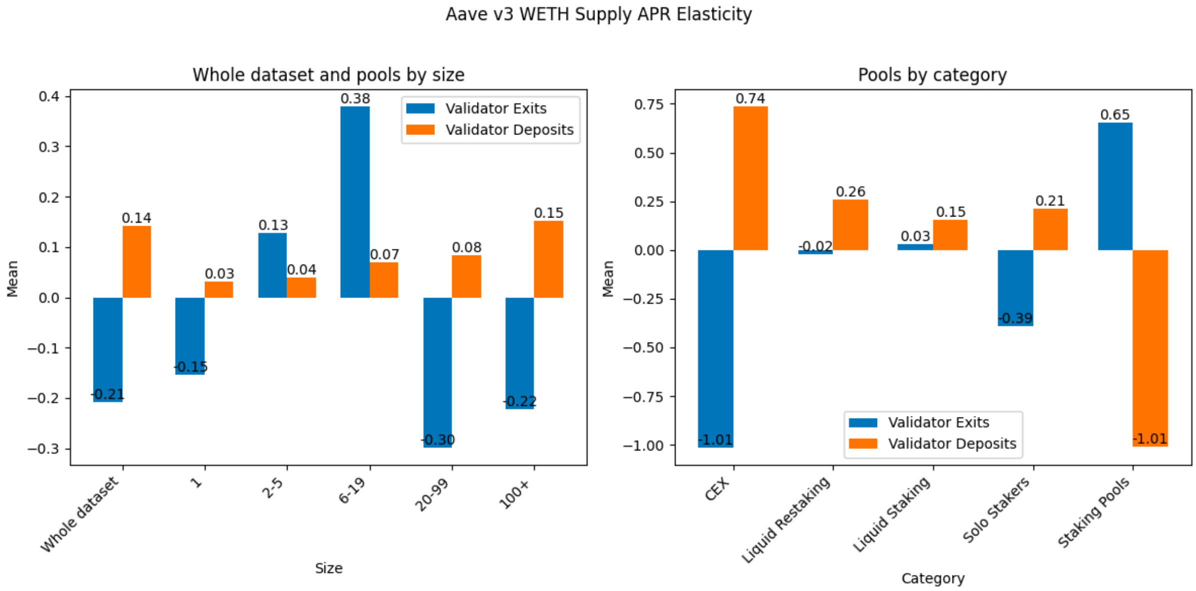

Figure C2: Validator exits and total count change relative to Aave v3 WETH supply APR. The amounts are normalized relative to subgroups.

Table C3

Curve stETH-WETH Liquidity APR Elasticity

This table reports elasticity estimates of validator exits, validator deposits, and the change in total validator count with respect to the Curve stETH-WETH Liquidity APR daily between slots 6,206,400 and 8,985,600. Panel A reports the mean elasticity, t-static, and standard deviation for the whole dataset as well as by different pool sizes. Panel B reports the same values grouped by different categories of staking entities. Panel C reports the same values for the largest individual staking pools. ***, **, and * denote statistical significance of two-sample t-test at the 1%, 5%, and 10% levels against a null hypothesis of no elasticity difference compared to the whole dataset.

Panel A. Whole dataset and pools by size

	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
Whole dataset	0.1266	-	2.6925	0.001	-	0.0433	0.0665	-	2.3916	387
1	0.2182	0.37	4.0394	0.0014	0.16	0.0285	-0.1807	-0.93	4.6424	387
2-5	0.1609	0.20	2.102	0.0024	0.48	0.0384	-0.2801	-1.56	3.65	387
6-19	0.0169	-0.66	1.8142	0.0019	0.26	0.0534	-0.0129	-0.40	3.1066	387
20-99	1.0467	0.86	20.8397	0.0008	-0.08	0.0091	-1.3049	-1.32	20.2786	387
100+	0.0817	-0.27	1.9394	0.001	0.00	0.0461	0.1426	0.37	3.2961	387

Panel B. Pools by category

	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
CEX	0.0577	-0.46	1.1206	0.0012	0.05	0.06	-0.1034	-1.15	1.654	387
Liquid Restaking	-0.0076	-0.98	0.1493	0.0268	1.37	0.3677	12.9867	1.17	217.5147	387
Liquid Staking	-0.0819	-1.29	1.6898	-0.0003	-0.31	0.0716	0.2465	0.83	3.5507	387
Solo Stakers	0.0787	-0.32	1.1593	0.0045	0.71	0.0873	-0.1044	-1.26	1.1847	387
Staking Pools	-0.0771	-1.25	1.7511	0.0058	0.89	0.0968	0.0371	-0.13	3.7352	387

Panel C. Largest pools

	Validator Exits			Validator Deposits			Validator Total Count Change			
Size	Mean	t(Mean)	SD	Mean	t(Mean)	SD	Mean	t(Mean)	SD	N
Lido	-0.1035	-1.36	1.9416	-0.0005	-0.32	0.0807	-0.0035	-0.43	2.112	387
Coinbase	0.0852	-0.27	1.3842	0.0045	0.58	0.1102	-0.0155	-0.48	2.403	387
Binance	-0.0958	-1.37	1.6979	0.0003	-0.31	0.0062	0.0654	-0.01	2.0653	387
Rocketpool	0.0351	-0.61	1.1591	0.0008	-0.07	0.0496	1.1243	1.31	15.7073	387
Kraken	0.2545	0.43	5.125	0.0007	-0.12	0.0091	-0.2991	-1.18	5.6093	387
OKX	0.1886	0.28	3.357	0.0007	-0.13	0.0131	-0.6449	-1.42	9.5604	387
Bitcoin Suisse	0.2126	0.38	3.5883	0	-0.44	0.0005	-0.2858	-1.53	3.8548	387
Ledger Live	0.0374	-0.64	0.4942	0.0063	0.71	0.1398	-0.3105	-0.84	8.4885	387
Ether.Fi	0	-0.92	0	-0.004	-0.84	0.0814	48.8121	1.07	676.9896	219
Mantle	3.4748	0.98	48.5938	0.0003	-0.32	0.0044	-4.0207	-1.16	50.2721	204
Other Stakers	0.3703	0.64	7.0348	0.0012	0.07	0.0304	-0.3065	-1.43	4.5327	387

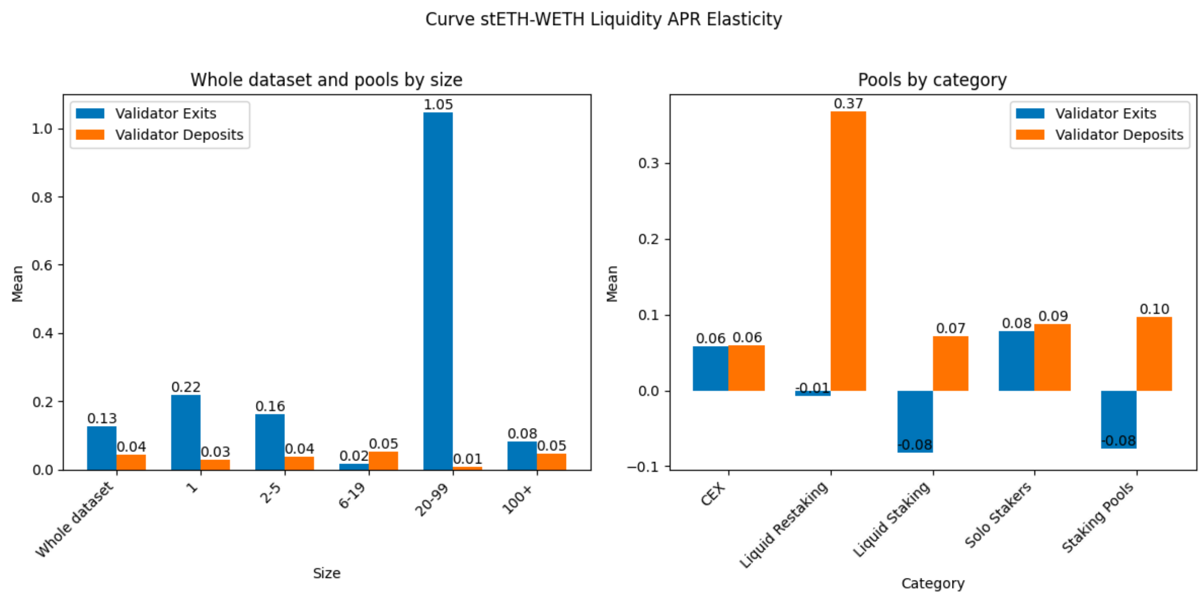

Figure C3: Validator exits and total count change relative to Curve stETH-WETH liquidity APR. The amounts are normalized relative to subgroups.

D. Event studies

Table D1						
Rocket Pool Mainnet Launch						
This table reports the results of an event study analyzing the impact of the Rocket Pool Mainnet launch (slot 2,466,000) on validator deposits. The study uses one-week (50,400 slots) pre-event and post-event windows and calculates the values on an hourly level. Panel A presents the mean and standard deviation of the result for the full dataset and across different pool sizes, both before and after the event. Panel B reports the same metrics grouped by different staking entity categories. ***, **, and * denote statistical significance of two-sample t-test at the 1%, 5%, and 10% levels against a null hypothesis of no elasticity difference compared to the whole dataset.						
Panel A. Whole dataset and pools by size						
	Validator Deposits					
Size	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	N
Whole dataset	0.2548	1.4151	0.1122	0.367	-	336
1	0.1754	1.0944	0.0984	0.4397	1.30	336
2-5	0.1333	0.5096	0.1154	0.3958	0.38	336
6-19	0.9198	9.9784	0.1645	0.8117	0.77	336
20-99	0.0675	0.5291	0.1162	0.7107	-0.88	336
100+	0.2481	1.3135	0.11	0.397	-0.28	336
Panel B. Pools by category						
	Validator Deposits					
Size	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	N
CEX	0.3195	1.6385	0.1714	0.6762	0.49	336
Liquid Restaking	-	-	-	-	-	0
Liquid Staking	0.2922	3.7477	0.0287	0.1682	-0.62	336
Solo Stakers	0.187	1.5944	0.1386	0.7611	-0.15	336
Staking Pools	0	0	0.0183	0.2014	-1.40	336

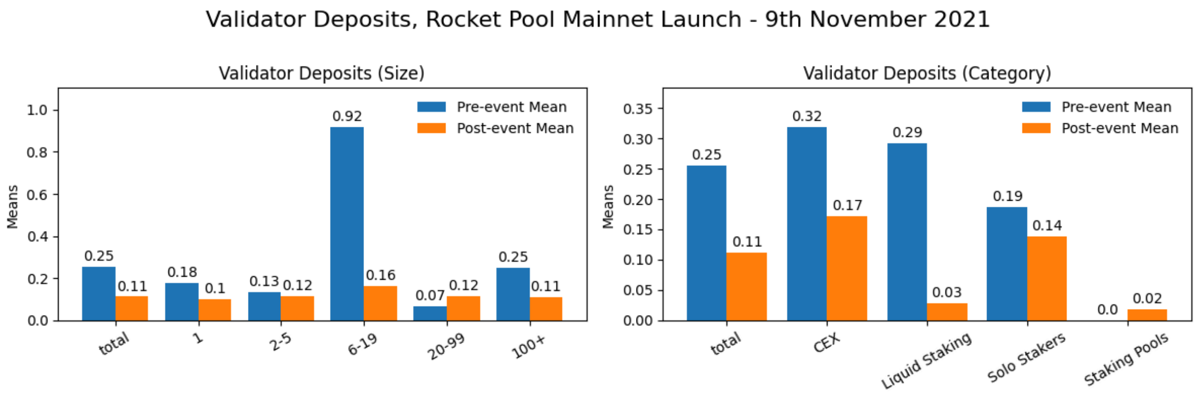

Figure D1: Validator deposits one week before and after Rocket Pool mainnet launch. The deposits have been normalized by the net deposits of the subgroup.

Table D2

First FED 0.75% Rate Hike

This table reports the results of an event study analyzing the impact of the first FED 0.75% rate hike (slot 4,042,800) on validator deposits. The study uses one-week (50,400 slots) pre-event and post-event windows and calculates the values on an hourly level. Panel A presents the mean and standard deviation of the result for the full dataset and across different pool sizes, both before and after the event. Panel B reports the same metrics grouped by different staking entity categories. ***, **, and * denote statistical significance of two-sample t-test at the 1%, 5%, and 10% levels against a null hypothesis of no elasticity difference compared to the whole dataset.

Panel A. Whole dataset and pools by size						
	Validator Deposits					
Size	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	N
Whole dataset	0.1105	0.5944	0.0903	0.3435	-	336
1	0.0711	0.2864	0.0506	0.2134	-0.77	336
2-5	0.0791	0.3326	0.1383	0.5297	-1.23	336
6-19	0.0821	0.4884	0.0688	0.3269	-0.41	336
20-99	0.1024	0.7906	0.0554	0.2544	0.06	336
100+	0.113	0.6515	0.0925	0.3704	0.66	336
Panel B. Pools by category						
	Validator Deposits					
Size	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	N
CEX	0.2408	1.5392	0.1547	0.7567	0.98	336
Liquid Restaking	-	-	-	-	-	0
Liquid Staking	0.0193	0.0755	0.0206	0.0864	-0.23	336
Solo Stakers	0.0962	0.5656	0.4518	3.7664	-0.87	336
Staking Pools	0	0	0	0	-	336

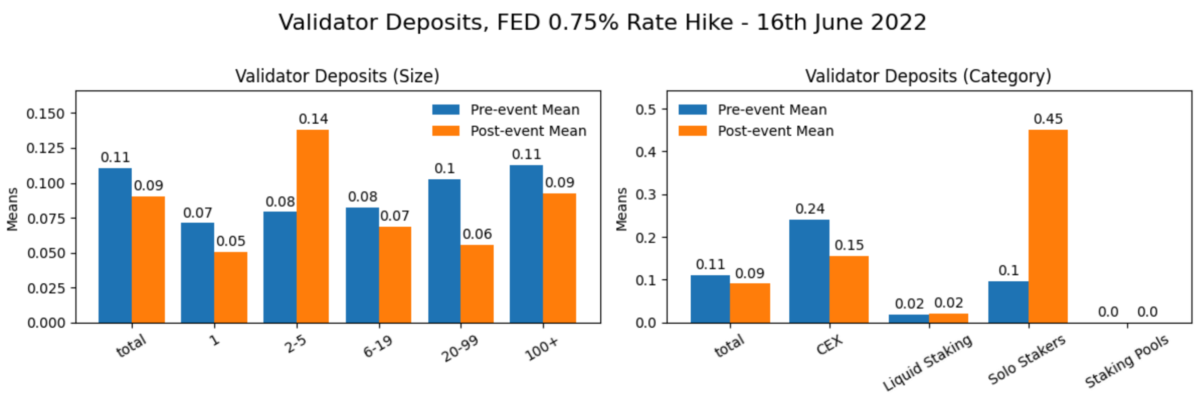

Figure D2: Validator deposits one week before and after the first 0.75% rate hike of the Federal Reserve. The deposits have been normalized by the net deposits of the subgroup.

Table D3						
Ethereum Staking Withdrawals Open						
This table reports the results of an event study analyzing the impact of the Ethereum staking withdrawals opening (slot 6,202,800) on validator deposits. The study uses one-week (50,400 slots) pre-event and post-event windows and calculates the values on an hourly level. Panel A presents the mean and standard deviation of the result for the full dataset and across different pool sizes, both before and after the event. Panel B reports the same metrics grouped by different staking entity categories. ***, **, and * denote statistical significance of two-sample t-test at the 1%, 5%, and 10% levels against a null hypothesis of no elasticity difference compared to the whole dataset.						
Panel A. Whole dataset and pools by size						
	Validator Deposits					
Size	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	N
Whole dataset	0.1059	0.2919	0.529	1.6221	-	336
1	0.1293	1.0617	0.2239	0.6801	1.19	336
2-5	0.0677	0.2666	0.3332	0.9785	0.04	336
6-19	0.1177	0.5851	0.5543	1.6367	-0.67	336
20-99	0.1001	0.4286	0.8990**	3.6011	-2.07	336
100+	0.1062	0.297	0.5160*	1.6456	1.88	336
Panel B. Pools by category						
	Validator Deposits					
Size	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	N
CEX	0.1445	0.4588	0.3882***	1.5009	3.53	336
Liquid Restaking	-	-	-	-	-	0
Liquid Staking	0.0828	0.2841	0.311	1.1248	0.13	336
Solo Stakers	0.0141	0.0761	0.4123**	2.0632	-2.44	336
Staking Pools	0.0706	0.4228	0.2953	0.9209	0.46	336

Table D4											
Bitcoin ETF Launch											
This table reports the results of an event study analyzing the impact of the Bitcoin ETF launch (slot 8,175,600) on validator deposits and exits. The study uses one-week (50,400 slots) pre-event and post-event windows and calculates the values on an hourly level. Panel A presents the mean and standard deviation of the result for the full dataset and across different pool sizes, both before and after the event. Panel B reports the same metrics grouped by different staking entity categories. ***, **, and * denote statistical significance of two-sample t-test at the 1%, 5%, and 10% levels against a null hypothesis of no elasticity difference compared to the whole dataset.											
Panel A. Whole dataset and pools by size											
	Validator Exits					Validator Deposits					
Size	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	N
Whole dataset	0.0123	0.0037	0.0065	0.0063	-	0.2195	0.6904	0.2361	0.6867	0	336
1	0.0019	0.0085	0.0065***	0.0143	-4.11	0.2331	0.7646	0.1888	0.5538	0.81	336
2-5	0.002	0.011	0.0077***	0.0189	-3.57	0.1974	0.5978	0.2052	0.6242	0.64	336
6-19	0.001	0.0053	0.0057***	0.0166	-2.96	0.2119	0.7318	0.198	0.7447	1.64	336
20-99	0.0022	0.0134	0.0057**	0.0225	-2.20	0.5159	4.3083	0.1574	0.8221	1.00	336
100+	0.0133	0.0042	0.0066***	0.0066	6.48	0.2056	0.6566	0.2416*	0.7181	-2.04	336
Panel B. Pools by category											
	Validator Exits					Validator Deposits					
Size	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	Mean Pre-Event	SD Pre-Event	Mean Post-Event	SD Post-Event	t(Mean)	N
CEX	0.0012	0.0056	0.0102***	0.0158	-9.35	0.1278	0.4419	0.1589	0.542	-0.77	336
Liquid Restaking	0	0	0	0	-	6.612	6.612	3.5688**	14.0832	2.40	336
Liquid Staking	0.0002	0.001	0.0021***	0.007	-3.55	0.1503	0.8737	0.0938*	0.5747	1.87	336
Solo Stakers	0.0004	0.0042	0.0027**	0.0144	-2.59	0.0546	0.5662	0.0248	0.1457	0.81	336
Staking Pools	0.2178	0.0922	0.0057***	0.021	24.63	0.7776	3.8646	1.3224	7.5208	-0.43	336